



Exploring the potential of functional photocatalytic nanomaterials for harmful algal blooms mitigation: a comprehensive review

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Abstract Harmful algal blooms (HABs) are one of such unprecedented issues that significantly disturb both freshwater and seawater and have posed a severe risk to human health, ecological security, and socio-economic growth. The present review primarily emphasises on the significant development on the extenuation of HABs using photocatalytic nano-architectonics. The sources and types of numerous algal toxins, their ecological influences, and health risk assessment are discussed. Further, the potential of various advanced functional photocatalytic nano-architectonics such as MOFs, TiO₂-based metal

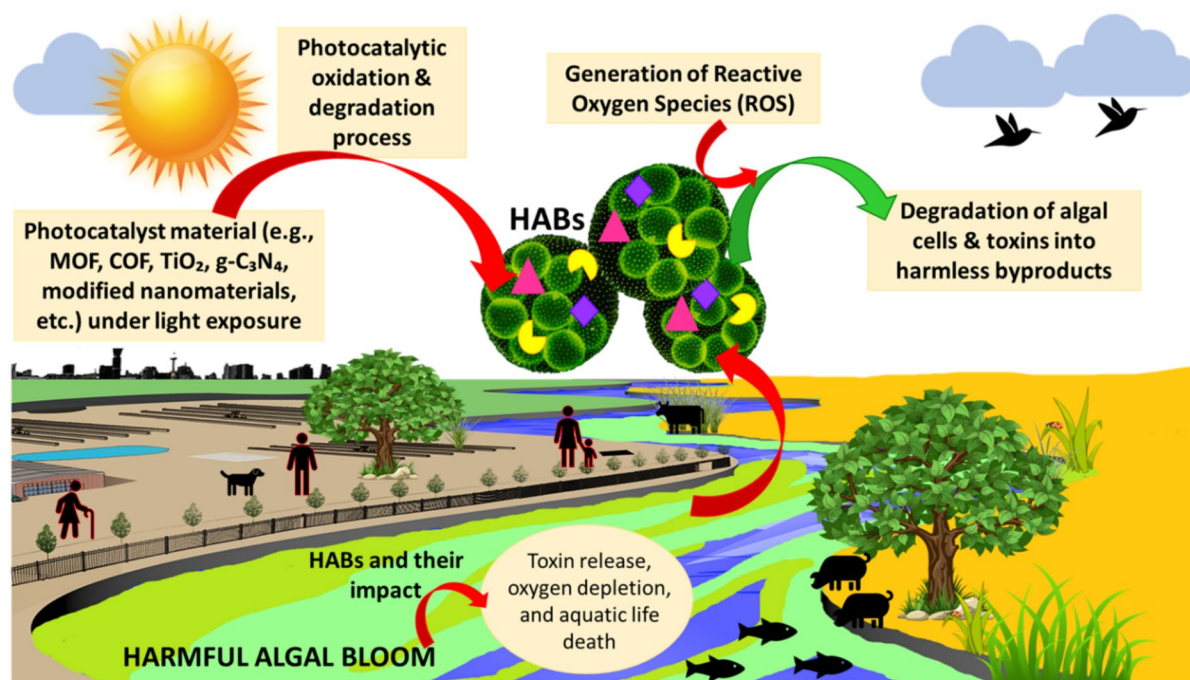
composites, carbon nitrides, COFs, graphene-based materials, zeolitic imidazolate frameworks, porous coordination networks, hybrid composite materials involving self-assembled metal ions/clusters with bridging organic ligands, biochar, etc., for HABs inhibition is explored. Furthermore, the plausible mechanisms of photocatalysts for the obliteration of HAB cells, along with numerous factors affecting the photocatalytic algal inactivation, are also covered. Finally, the review summarizes the discussion on the limitations, challenges, future perspectives, and solutions for further research in this research area.

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Graphical abstract



Keywords Cyanotoxins · Human health risk · Metal organic framework · Covalent organic framework · Graphene · g-C₃N₄

1 Introduction

Globally, harmful algal blooms (HABs) are one of the perilous ecological abnormal phenomena that have a direct influence on water quality and public health issues (Moumen et al. 2022; Kim et al. 2023a; Gao et al. 2024a; Mohammed and Arockiaraj 2024; Zheng et al. 2024). The HABs epitomize the excessive growth of photoautotrophic algae/cyanobacteria that form dense biomass and produce toxic secondary metabolites (cyanotoxins) that lead to the annihilation of ecosystem structure and function (Wang et al. 2023c; Li et al. 2024). HAB results in contaminating water, intimidating microbial diversity, and instigating antagonistic effects on human health and livestock animals, and has an annual economic impact projected to cost tens of billions of USD (Sik Choi et al. 2022; Wang et al. 2023d; Kim et al. 2024; Zheng et al. 2024; Zhu et al. 2024). Around

60% of China's lakes are facing particularly acute HAB issues, and the proportion of lakes affected by blooms continues to expand further into high-latitude and oligotrophic lakes (Huisman et al. 2018; Park et al. 2024). Since 1998, the World Health Organization has recommended a drinking water guideline of 1 ppb for microcystin-LR (He et al. 2021; Yang et al. 2023c).

In this regard, there was a critical prerequisite to detect, control, and resolve the HAB epidemics at the earliest. Several conventional HAB removal techniques, such as physical, chemical, and biological, are classified (Zhu et al. 2021; Sun et al. 2023a; Marone et al. 2024). Each of these techniques has several advantages and disadvantages that cannot be disregarded as well. Photocatalysis is an emerging field compared to traditional HAB removal techniques (Li et al. 2025; Wang et al. 2023a, 2023b, 2023c). Over the past decade, there has been a rising trend in this area due to its promising potential and efficiency, stability, and wide range of applications (Li et al. 2022a, b, 2025; Sheik et al. 2024; Rolim et al. 2023; Wang et al. 2023a; Shi et al. 2025). In comparison to the traditional methods, the photocatalytic technique has

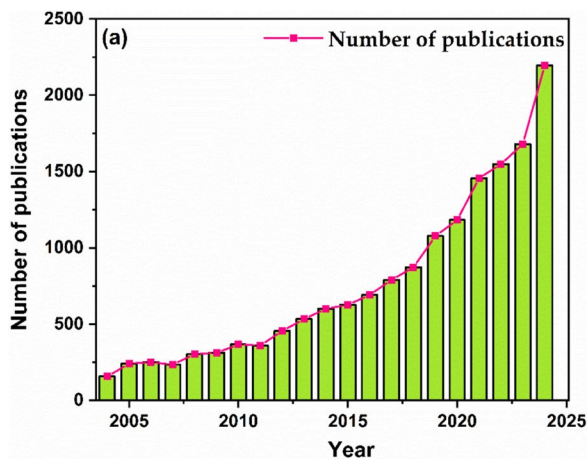
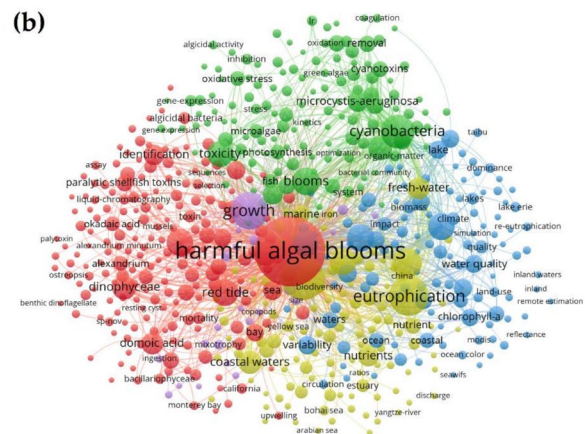


Fig. 1 a Representation of the year wise publications/research outputs in association with the HABs [till date] from 2004 to 2024 in ScienceDirect and **b** scientometric visualization of the

gained extensive credit for regulating and resolving HABs from the aqueous bodies. Here, Fig. 1a represents the year-wise publications/research outputs in association with the HABs (till date) from 2004 to 2024 in ScienceDirect, and it can be seen that the keyword HABs has continuously flourished over the past two decades. In addition, Fig. 1b represents a scientometric visualization of the top 580 keywords (out of 27,099) within the “Web of Science Core collection,” using “HABs” as the prime keyword (assessed on 11 Feb, 2025) with 30 minimum number of occurrences.

Recently, there has been rising interest in advanced functionalized materials such as metal–organic frameworks (MOFs), carbon nitrides (CN), covalent organic frameworks (COFs), TiO₂-based metal composites, graphene-based materials, zeolitic imidazolate frameworks, polyoxometalates, coordination pillared-layer series materials, porous coordination network, coordination complexes of bipyridines, hybrid composite materials involving self-assembled metal ions or metal clusters with bridging organic ligands, etc. (Goswami et al. 2024, 2025a; Li et al. 2024, 2025; Shi et al. 2025). These abovementioned functional materials have considerable specific surface area, structural stability, high thermal and chemical stability, tunability, and in-situ modification, and have numerous applications in catalysis, surface chemistry, energy storage, gas sorption, wastewater treatment, and drug delivery systems (Wang et al. 2023a, c; Kushwaha et al. 2024; Li et al. 2025).



top 580 keywords [out of 27099] within the database of “Web of Science Core Collection,” using “HABs” as the search keyword [with 30 minimum number of occurrence]

Henceforth, the present review article focuses on the mitigation of HABs using advanced functional photocatalytic nano-architectonics. Initially, the different types and sources of various algal toxins, their ecological impacts, and health risk assessment are discussed, followed by the potential of several advanced functional photocatalytic nano-architectonics such as MOFs, COFs, biochar, graphene, zeolite, TiO_2 composites, $\text{g-C}_3\text{N}_4$, etc., for HABs inhibition are discussed in detail. The plausible mechanism of photocatalysts involved in the destruction of algae cells, along with the factors affecting the photocatalytic algal inactivation, was also covered. The significance and toxicity of post-photocatalytic transformed products for HAB removal are elaborated. Lastly, the discussion on the limitations, challenges, environmental and eco-toxicological risks posed by the photocatalysts, sustainability issues, role of AI/ML for designing the photocatalysts, process optimization of mitigating strategies, kinetic modeling, and mechanistic inference of the algae and their toxin degradation, regulatory frameworks, and solutions for future research were proposed.

2 Algal toxins

2.1 Categories and algal species producing toxins

Lethal compounds and metabolites, viz., microcystins, saxitoxin, anatoxins, and others, cause

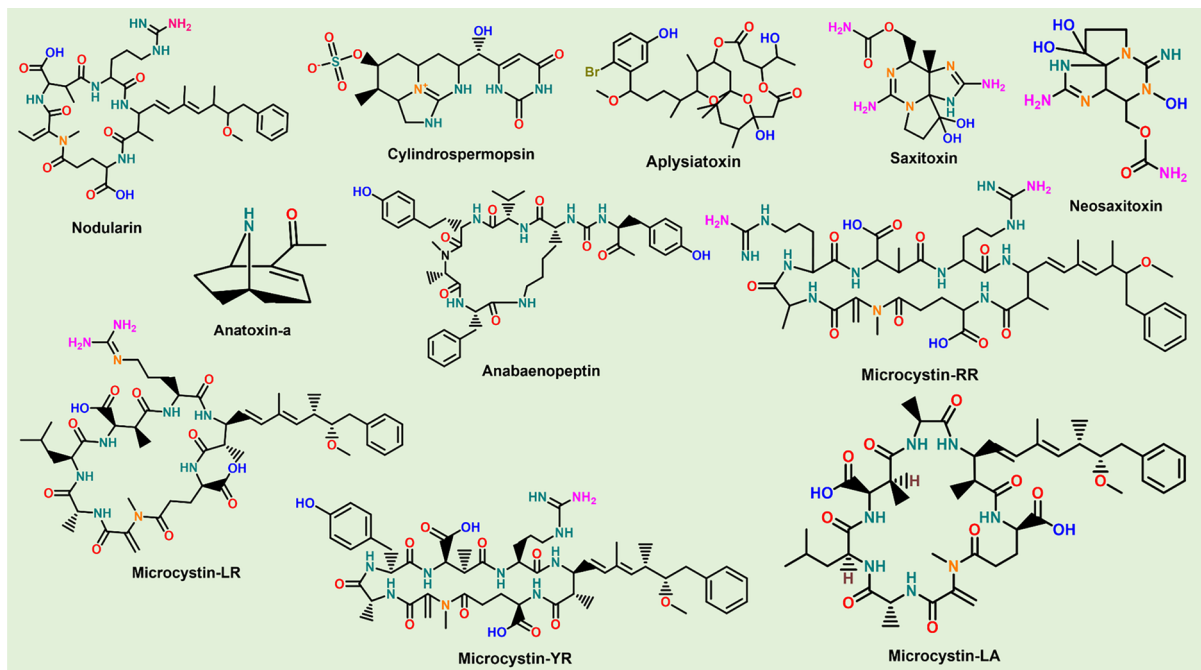


Fig. 2 Chemical structures of some commonly studied algal toxins

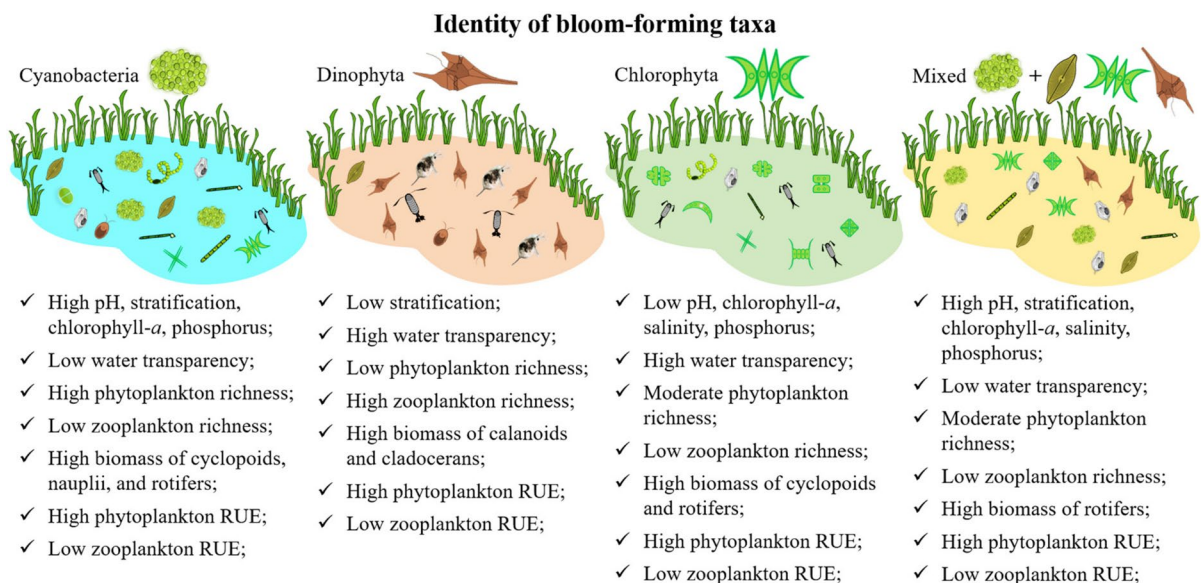


Fig. 3 A diagram representing the key ecological effects of algal blooms, affected by diverse bloom-forming phytoplankton taxa, on freshwater reservoirs (RUE: Resource Use Efficiency) (reprinted with permission from Amorim and Moura (2021))

considerable threats to inland and marine ecosystems (Nowruzi and Porzani 2021). However, acute human health consequences from algal toxins are moderately

uncommon; they can lead to gastroenteritis, liver disorders, kidney dysfunction, and even mortality in the most extreme circumstances. Different classes of

toxins include cyanotoxins (Sun et al. 2023c), dinoflagellate toxins, diatom toxins, and prymnesin toxins (Wagstaff et al. 2021). Recognizing and assessing these algal toxins in aquatic bodies is essential for confirming the protection of aquatic sources and for safeguarding human and animal health. Humans are most likely to come in contact with algal toxins during recreational activities in water bodies, frequently through incidental consumption or inhalation. Fig. 2 represents the chemical structures of some commonly studied algal toxins. Microcystin is the most widespread and has been correlated to human malignancies. Furthermore, domestic animals and livestock are threatened with lethal cyanotoxin poisoning. Insistent cyanobacterial blooms have also caused the fatalities of fish and birds, further emphasizing the environmental influence of these toxins (Ly et al. 2024).

The most prevailing and threatening algal species are diatoms, dinoflagellates, and cyanobacteria, with cyanobacteria as the most lethal and causing the most significant hazard to animals (Ly et al. 2024). The proliferation of algae is generally the outcome of surplus phosphorus and nitrogen in water bodies. It can also be established in new ecosystems through boats or cyclones that transfer them to diverse habitats. Fig. 3 represents the foremost environmental consequences of algal blooms, comprised of diverse bloom-forming phytoplankton taxa. The most commonly affected animals are birds during mass death events triggered by toxicoses, even though the illnesses and fatalities of wild and domestic fauna are intermittent but prevalent. Moreover, several municipal and recreational water sources have been associated with human poisoning incidents. HABs can trigger five evident diseases, comprising ciguatera seafood intoxication, as well as four shellfish intoxication syndromes: paralytic, amnesic, and diarrhoeic shellfish poisoning. Between these, ciguatera seafood poisoning and paralytic shellfish poisoning (Ben-Gigirey et al. 2020) are believed to be extremely dangerous. Provincial aspects such as increasing temperatures, augmented eutrophication encouraged by anthropogenic nutrient supplementation, and a rise in intense hydrological events (e.g., storms and droughts) promote the bloom of HABs from freshwater to aquatic environments (Paerl et al. 2018).

2.2 Aspects impelling HABs toxin production

Eutrophication and the correlated GHG releases are efficiently controlled via anaerobic digestion (Singh et al. 2023). Detecting and assessing the aspects prompting algal toxin production is an important concern that is influenced not only by the individual algal species but also by ecological conditions. To forecast and monitor algal toxin epidemics, these aspects must be cautiously identified and carefully supervised. Numerous key features affect algal toxin production, liable to the species and the exclusive attributes of the environment including: (i) nutrient accessibility (a surplus of nitrogen and phosphorus can substantially encourage toxin production) (Bonilla et al. 2023), (ii) water temperature (especially changes with the change in climate conditions, playing a fundamental part in toxin production), (iii) light (suitable light is vital for photosynthesis, directly influencing algal development and toxin production), (iv) salinity and water chemistry (alterations in salt and water chemistry can affect algal growth and production of the toxin), (v) pH level, (vi) competition and predation (cooperation with another algae and predators can influence toxin production) (Yang et al. 2023a), (vii) environmental intrusions (aspects like variations in river flow, storm incidents, and pollution from chemicals can interrupt environments and influence toxin production) (Zhang et al. 2018b). Knowing these interconnected issues is necessary for attaining effective approaches to alleviate algal toxin epidemics and safeguard aquatic environments.

2.3 Ecological impacts of HAB toxins

2.3.1 Effects of HAB toxins on aquatic bodies

HAB toxins have severe effects on aquatic environments. Globally, freshwater ecosystems are intensely susceptible to these toxins owing to their bio-accumulative characteristics. These toxins not only contaminate water bodies but also accumulate in aquatic flora, fauna, and sediments, eventually damaging all components of the biota (Pham and Utsumi 2018). Table 1 represents some of the ecological impacts of algal blooms. Protecting aquatic environments from algal toxins involves efficient detection and supervision. Preventive procedures, such as decreasing unnecessary nutrient inputs, observing water quality, and executing

Table 1 Ecological impacts of algal blooms

Ecological impacts due to HABs	Key Findings	Reference
Altered food web	Algal blooms disrupt the balance of aquatic food webs by favoring herbivorous zooplankton, impacting water quality and higher trophic levels as well	Lan et al. (2024)
Habitat deterioration	Algal growth near water surface inhibits seagrasses and submerged aquatic plant growth and affects aquatic habitats	Imai et al. (2021); Lu et al. (2022a, 2022b)
Eutrophication	Excessive nutrient loading fuels algal blooms, changing aquatic ecosystem composition and nutrient cycling	Ly et al. (2021); Xiao et al. (2024)
Biodiversity loss and alteration	HABs cause deoxygenation of seawater, negatively impact the diversity of aquatic species, and alter ecological dynamics	Brown et al. (2020)
Algal toxins bioaccumulation	Cyanotoxins produced during blooms threaten fish and aquatic life, leading to mass mortalities	Rolton et al. (2022)
Oxygen depletion (hypoxia and anoxia)	The death and decomposition of algal blooms result in consumption of huge quantity of oxygen dissolved in water. It leads to low oxygen levels resulting in extensive hypoxia and significant mortality of benthic biota and fish	Karim et al. (2002)
Decreasing water clarity and quality	HABs significantly reduce water quality by increasing pH, stratification, eutrophication, and reducing water transparency	Amorim and Moura (2021)
Reduced light penetration	Dense algal blooms block the sunlight from penetrating the water surface and significantly disrupt photosynthesis and impact aquatic plant growth	Gallegos and Jordan (2002)
Negative impact on shore birds	It reduced food availability and further indirectly altering their foraging behavior	Yan et al. (2022a, b)

primary precautionary systems, are vital to alleviating HAB and decreasing its effects on the environment and human activities (Marampouti et al. 2020).

2.3.2 Impact of HABs' toxins on biodiversity and food chains

Recently, lethal HAB blooms directed by ocean acidification have interrupted oceanic food webs, initiated cascading impacts through numerous trophic levels (Lu et al. 2022b), and presented problems to freshwater biodiversity conservation approaches (Ahmed et al. 2022). Furthermore, eutrophication, HABs, and biodiversity-changing aspects are further problematic due to complicated nutrient changes (Glibert 2017). Algal toxins can penetrate sediments, marine animals, and terrestrial and aquatic plants via contaminated

water, eventually contaminating the food web and presenting considerable health hazards (Abeyawardena et al. 2018; Patiño et al. 2023). Insight and mitigating the influences of algal toxins are necessary for maintaining the health and flexibility of marine environments. To preserve biodiversity and the food web, programs must emphasize lowering nutrient contamination, executing sustainable fishing methods, and implementing ecosystem-based methods. The biological and ecological effects of HABs toxins are widespread, affecting manifold levels of the food chain and threatening ecosystem stability.

2.3.3 Ecological impacts

HABs disturb aquatic ecosystems and have serious ecological repercussions since they are brought on by

the overgrowth of specific microalgae or phytoplankton. By outcompeting other organisms for nutrients, these blooms change the structure of the food web and take the lead as a food source for filter-feeding and herbivorous zooplankton. Fish and more advanced trophic levels may be impacted by the growth of particular zooplankton populations brought on by this imbalance (Amorim and Moura 2021). Still, the food chain is further upset by the fact that many algae species are poisonous or inedible to grazers. For example, more than 6,000 tons of farmed salmon perished in Chile's Los Lagos region due to a 2021 bloom of the raphidophyte *Heterosigma* (Mardones et al. 2023). Through the reduction of sunlight penetration caused by dense surface algae, HABs also physically change aquatic ecosystems. Fish and other animals that depend on submerged aquatic plants for food and habitat may suffer from slower growth or even extinction (Zhu et al. 2021). The health of humans and ecosystems is extremely vulnerable to the powerful toxins released by cyanobacterial HABs, often known as blue-green algae, viz., cylindrospermopsins, anatoxins, saxitoxins, and microcystins (Yang et al. 2022). Skin irritation, neurological symptoms, liver damage, and gastrointestinal problems can result from human exposure to cyanotoxins through tainted water (Nielsen and Jiang 2020). Additionally, these toxins affect aquatic creatures as well, raise the incidence of sickness, and induce mass mortality in fish (Zhang et al. 2023b). In general, HABs offer serious risks by producing toxins that affect human health and biodiversity in addition to upsetting aquatic ecosystems by changing food webs and habitats.

2.4 Health risks allied with HAB toxins

The most identified illnesses associated with HAB are paralytic, neurotoxic, and diarrhetic shellfish poisonings (Grattan et al. 2016). The mode and incidence of contact with algal toxins via ingestion, inhalation, or skin contact depend on the degree of bacterial contamination in the natural environment (Rogers and Stanley 2023). Environmental contaminants, involving algal toxins, enter organisms through several routes and are metabolized inside, possibly triggering damage (Neves et al. 2021). These toxins are extremely harmful and allied to an extensive range of health problems involving neurological, dermatological, pulmo-no-toxic, hepatotoxic, and immuno-toxic

disorders, besides cancer (Faulkner et al. 2025). However, assessing the full scope of algal toxins' harmful impacts on human continues to be challenging due to gaps in existing data. Algal toxins lead to considerable health hazards, specifically through contamination of food chains and water. Major health concerns comprise the consumption of contaminated seafood (shellfish accumulate algal toxins during HABs can trigger acute health concerns when eaten) (Bechard and Lang 2023), consumption of algal toxin-contaminated freshwater (Martino et al. 2020), algal toxins can develop into airborne when water containing toxins splashes or disruptions into droplets and spreads in the air (Hu et al. 2020), and finally, water bodies polluted with algal toxins can disturb communities dependent on them for drinking water, leisure, and fishing (Martino et al. 2020). Addressing these health, economic, and social challenges demands effective administration, mitigation approaches, and public understanding to lower contact with algal toxins and their harmful effects.

3 Photocatalytic growth inhibitor of cyanobacteria

Earlier, numerous physicochemical and biological treatment technologies were investigated for eliminating the HAB cells and toxins are extensively employed owing to their effortlessness and prompt efficiency. But they can also cause involuntary consequences, for instance, cell damage, lysis, and toxin release. Substantial investigation attempts have been dedicated for improving water treatment methods to enhance the exclusion of cyanobacteria and cyanotoxins while reducing these disadvantages. Table 2 represents the comparison of various methods (physical, chemical and biological) for HABs control. Chemical treatments are employed mostly where the biological treatment techniques fail. Further, biological treatment is well-known for its eco-friendly and cost effectiveness for HAB removal from wastewater (Yan et al. 2023; Ramírez-Escárcega et al. 2025). Recently, photocatalytic methods using advanced functional materials are very much in trend and have opened new dimensions for HAB removal in an outstanding way. Therefore, the current section represents an outline on the significant progress on the mitigation of HABs using photocatalytic architectonics along with

Table 2 Comparison of various methods for HABs control

Methods	Techniques	Advantages	Limitations	References
Chemical methods	Metals (aluminum, iron, copper, calcium)	Low cost High residence time	Noxiousness against non-target species Accretion in the environment	Magdaleno et al. (2014)
	Photosensitizers (hydrogen peroxide, phthalocyanines, titanium dioxide)	Inexpensive Degradability	Dangerous manipulation Coloration	Pohl et al. (2015)
	Herbicides (diuron, endothall, atrazine, simazine)	Inexpensive High residence time	Noxiousness Release of toxins	Nagai et al. (2016)
Physical methods	UV irradiation	Environmental friendly Pollution free	High energy requirement To be confirmed at up-scaled levels	Zamir Bin Alam et al. (2001)
	Membrane filtration technology	Well-established technology High stability	High cost	Zhao et al. (2017)
	Adsorption	Eco-friendly Pollution free	High cost To be confirmed at up-scaled levels	Marzbali et al. (2017)
Biology methods	Aquatic plants	Technically simple reactor	Affect biodiversity Deteriorate eutrophication	Montemezzani et al. (2017)
	Aquatic animals	User-friendly Environmentally sound	It will not work in oxygen-poor conditions Affect biodiversity Poor efficiency	Montemezzani et al. (2017)
Combined technologies	Microorganisms	High specificity High efficiency	High cost To be confirmed at up-scaled levels	Sun et al. (2016)
	Ultrasonic radiation and jet circulation to flushing	High efficiency	High cost To be confirmed at up-scaled levels	Nakano et al. (2001)
	Combination of uniform design with artificial neural network coupling genetic algorithm	High efficiency Low cost	To be confirmed at up-scaled levels	Cai et al. (2014)

the plausible mechanisms and factors affecting the removal techniques are covered.

The semiconductors with broad bandgaps have difficulty in exploiting visible light effectively. To overcome these drawbacks, scientists have created heterojunction structures that accelerate electron movement and improve charge separation. The widely explored heterojunctions include Type II and Z-scheme, both of which increase charge carrier separation and electron transport effectiveness (Fig. 4). The charge separation by Type II heterojunctions advances via electrons and holes shifting to lower energy levels,

however, this method deteriorates the redox capability of photocatalyst.

On the other hand, Z-scheme heterojunctions uphold effective redox capability by recombining the low-energy electrons and holes while saving the high-energy transporters. This process permits Z-scheme heterojunctions to demonstrate excellent photocatalytic functioning, compelling them to be the ideal alternative for photocatalytic treatments. Additionally, element doping is an alternative efficient approach for increasing photocatalytic effectiveness, remarkably for materials with broad band gaps. The conduction band (CB) and valence band (VB)

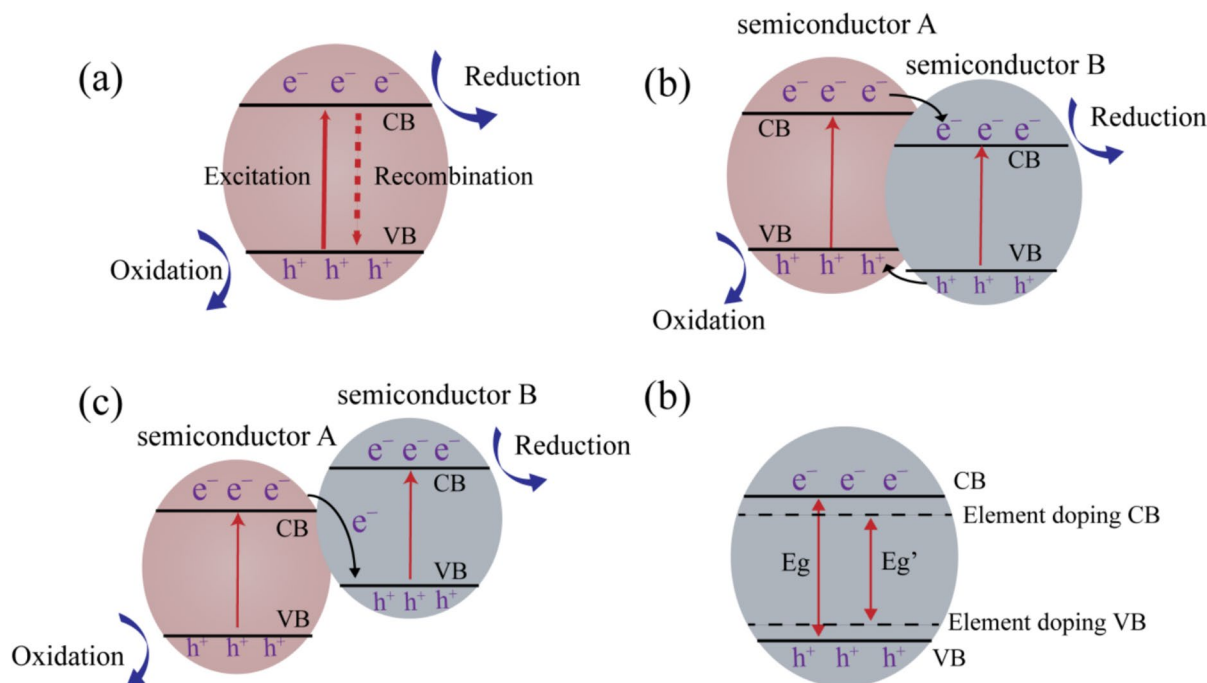


Fig. 4 Photocatalytic mechanism of single semiconductor; **b** charge transfer pathway of Type II heterojunctions; **c** charge transfer pathway of Z-scheme heterojunctions and **d** mechanism of non-metallic element doping (reprinted with permission from Sun et al. (2023a))

energy levels can be modified by doping with non-metallic elements, which successfully lowers the bandgap and boosts the adsorption of visible light. These alteration approaches, involving heterojunction engineering and element doping, have been extensively employed in photocatalytic studies to degrade pollutants, substantially improving the efficacy of algae deactivation methods (Natarajan et al. 2018b; Xu et al. 2020).

3.1 Titania-based nanoparticles

Titanium dioxide (TiO_2) has been significantly examined amidst all semiconducting materials because of its physical and structural properties. F-Ce co-doped with TiO_2 was immobilized on expanded perlite (EP) using a sol-gel approach for photocatalytic elimination of algal cells (Wang et al. 2017a). The catalyst (F-Ce- TiO_2 /EP450) presented a honeycomb-like formation, and 350–650 °C calcination temperatures influenced electronic properties, doping sites, and surface area. After 9 h of solar-light treatment, it accomplished 98.1% algal elimination via ROS production, which interrupted

the photosystem I (PS I) and photosystem II (PS II) systems and damaged the dissolved organic matter present both extracellular and intracellular. Furthermore, microcystin-LR (MC-LR) was also removed (91%) during the process. The recovery was simple due to the catalyst's floating ability, with three cycles of stability and reusability. EP's permeability participated in ~56% algal adsorption, and 98% algal degradation employing F-Ce- TiO_2 /EP450, emphasizing the key function of adsorption. But photocatalytic effectiveness under varying settings (pH, light intensity, catalyst dosage) was not assessed.

Fe_2O_3 - TiO_2 nanoparticles produced by the ultrasonic-aided co-precipitation process employing TiCl_4 and $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, exhibited anti-algal activity towards *Chlorella vulgaris* (Baniamerian et al. 2018). The cellular viability was evaluated by assessing the chlorophyll content, exposing that light strength and photocatalyst quantity substantially affected chlorophyll 'a' breakdown. In freshwater, the highest deactivation (98%) ensued at 0.25 g/L, 55 W/m^2 visible light in 24 h. Furthermore, salinity affected NP effectiveness due to $\cdot\text{OH}$ scavenging

and active site blockage by halide ions. Although 99% algal elimination was completed, 24 h is impossible for high-load contaminated water, compelling improved amendments. Furthermore, the findings required discussion on recovery of catalyst, reusability, and separation, which are vital for field purposes. The imminent investigation should emphasize improving effectiveness, lowering treatment interval, and upgrading catalyst sustainability.

Natarajan et al. (2018a, b) used chitosan nanocomposites (TiO_2/Ag) and pristine (TiO_2 and Ag) films to inhibit *Dunaliella salina* under UV-C. Findings showed chitosan nanocomposites degraded glycocalyx formation and higher biofilm inhibition compared to pristine films. The cell viability was reduced, and membrane destruction was observed due to the ROS production and Ag^+ leaching, thus increasing the anti-fouling efficacy. Wang et al. (2017b) developed N, P co-doped TiO_2 /expanded graphite carbon layer (NPT-EGC) composites via sol-carbonization to inhibit *M. aeruginosa*. The photocatalyst NPT-EGC450 demonstrated superior photocatalytic activity, accomplishing 98.15% *M. aeruginosa* elimination in 9 h when exposed to visible light.

The mechanism involved quick adsorption, subsequently decreasing cell viability. After three cycles, the catalyst efficiently reduced algal cells and MC-LR, with 90% effectiveness, representing tremendous reusability and stability. In eutrophic lakes, *M. aeruginosa* flourishes efficiently and releases lethal MC-LR. To prevent this, Wang et al. (2018a) synthesized

PDDA-modified TiO_2 floating photocatalysts. This composite presented optimum performance, enhancing adsorption from 70.4% to 88.9% in the dark and total elimination from 77.8% to 92.6% during visible light exposure after 2 h. MC-LR degradation (96.55%) was observed within 3 h, destroying its Adda side chain. This catalyst displayed a strong ability for the removal of algae and toxins in eutrophic waters.

3.2 Non-metal doped TiO_2

Highly efficient TiO_2 is very efficient in photocatalysis for the inactivation of HAB in the light. Ivandic et al. (2022) exemplified the dispersion and immobilization of photocatalytic TiO_2 nanoparticles for cyanotoxin removal from water. Comparison of different concentrations of TiO_2 coated and uncoated nylon net supports for determining Microcystic-MC-LR degradation were evaluated (Fig. 5). Five-times coated TiO_2 (0.25 g/L dispersion) showed net 96%, 64%, and 10% removal for UV, indirect UV, and no UV exposure over 2 h and followed pseudo first-order kinetics. Wang et al. (2019) prepared TiOX ($X = \text{N}$ and P) photocatalyst via a facile sol-gel technique. The oxygen-deficient environment enhanced the doping of N and P into TiO_2 crystals by controlling the calcination temperature. Here, TiOX prepared at 550 °C displayed the best photocatalytic efficiency with *M. aeruginosa* inactivation rate of 81.5% in 6 h under visible light.

Fig. 5 In-situ approach to HABs degradation of microcystin-LR cyanotoxins using immobilized TiO_2 photocatalysts nylon net (reprinted with permission from Ivandic et al. (2022))

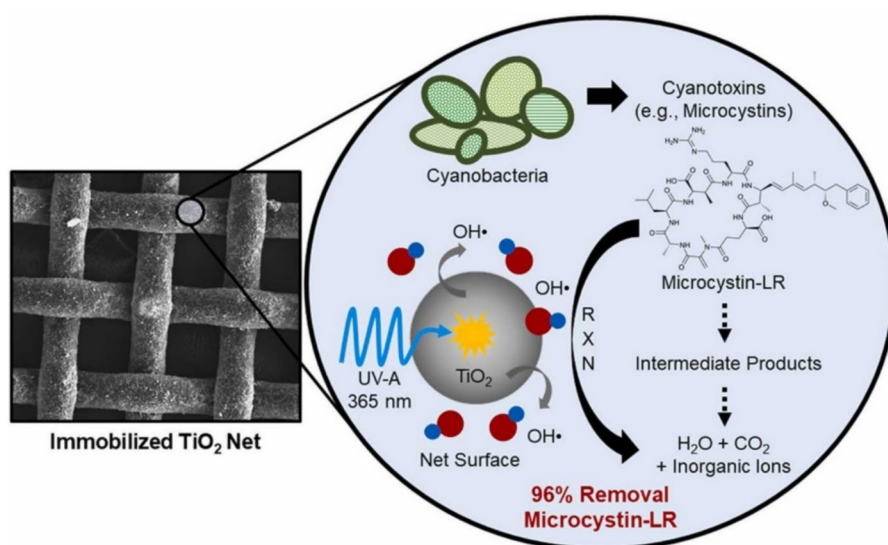


Table 3 Prevailing research on photocatalytic inactivation of *M. aeruginosa*

Photocatalysts	Light source	Time (h)	Concentration of algae cells (cells/mL)	Catalyst dosage (g/L)	Removal rate (%)	References
TiOX (X=N and P)	Vis	6	2.7×10^6	2	81.5	Wang et al. (2019)
TiO ₂ /g-C ₃ N ₄	Vis	6	2.7×10^6	2	88.1	Song et al. (2018)
Ag ₃ PO ₄ /b-N-TiO ₂	Vis	8	3×10^6	0.2	95	Zhou et al. (2022)
g-C ₃ N ₄ /MoO ₃	Vis	3	2.7×10^6	0.3	97	Wang et al. (2020)
Ag ₂ O/g-C ₃ N ₄	Vis	6	4.78×10^6	0.05	99.94	Fan et al. (2020a)
Ag ₂ CO ₃ -N:GO	Vis	5	4.78×10^6	—	100	Fan et al. (2020a)
Ag ₃ PO ₄ @ polyaniline	Vis	3	5.16×10^6	0.02	99.2	Fan et al. (2022a)
ZnFe ₂ O ₄ /Ag ₃ PO ₄ /g-C ₃ N ₄	Vis	3	5.16×10^6	0.1	96.33	Fan et al. (2021b)
Zn/Fe-LDHs	Vis	3	2.8×10^5	0.3	80	Gu et al. (2016)
Cu ₂ O-montmorillonite	Vis	3	2.8×10^7	0.4	90.4	(Gu et al. 2016)
Ag/AgCl@ ZIF-8	Vis	6	4.78×10^6	0.01	93.1	Fan et al. (2019c)
Ag/AgCl@ LaFeO ₃	Vis	3	6×10^9	0.05	100	Fan et al. (2021a)
Bi ₂ Mo ₃ O ₁₂	Vis	6	1.8×10^7	1	84	Li et al. (2018)
SnO ₂	UV light	1	2.8×10^8	0.3	95	Gu et al. (2018)
Ag-TiO ₂	Solar light	24	2.5×10^5	2.1	92	Chang et al. (2015)
CeO _x /TiO _{2-y} F _y	Vis	4	2.7×10^6	2	100	Wang et al. (2018a)
BiO _{2-x} /Bi ₃ NbO ₇	Vis	2.5	2.6×10^6	0.3	91	Xu et al. (2022)
g-C ₃ N ₄ @ Bi ₂ MoO ₆ @ AgI	Vis	6	4.86×10^6	—	99.9	Sun et al. (2022b)

3.3 Metal doped TiO₂ and composites

Cao et al. (2024) synthesized a Z-scheme AgBr/NH₂-MIL-125(Ti) composite with exceptional visible light absorption for efficient *M. aeruginosa* inactivation. The degradation efficacy of the composite for chlorophyll *a* was 98.7% after 3 h of visible light. Also, the removal rate was maintained at 91.1% after 5 times of repetitive utilization. Here, superoxide radicals ($\bullet\text{O}_2^-$) were the dominant ROS involved. By loading the photocatalyst on magnetic/floatable carriers is one of the prevalent techniques for overcoming the non-floatability and difficulty in recycling the photocatalysts' inadequacies. Table 3 represents some existing studies available on the photocatalytic inactivation of *M. aeruginosa*. Tian et al. (2025) synthesized CdZnS/TiO₂ membrane photocatalyst with adaptable suspended depth (during floating) and flexible assembly that have better applicability.

3.4 BiVO₄/TiO₂/NaY-zeolite ternary composite

Several studies have investigated metallic and non-metallic compounds for reducing environmental contaminants, involving cyanotoxins (Nawaz et al. 2018; Kim and Lee 2021). Lately, Ebrahimi et al. (2021) synthesized an innovative ternary BiVO₄/TiO₂/NaY-zeolite nanocomposite using a hydrothermal method followed via calcination for 6 h to degrade MC-LR. Qualitative analysis has established its crystalline structure, mineral morphology, and visible light absorption. Results showed effective removal (98%) of MC-LR by a type-II mechanism, with $\bullet\text{OH}$ production as the main reactive species at BiVO₄. Zeolite functioned as a mediator, increasing MC-LR interaction and inhibiting charge recombination in the ternary link. Though zeolites are substantially porous minerals comprised of silicon and aluminum, their adsorption competence was not assessed.

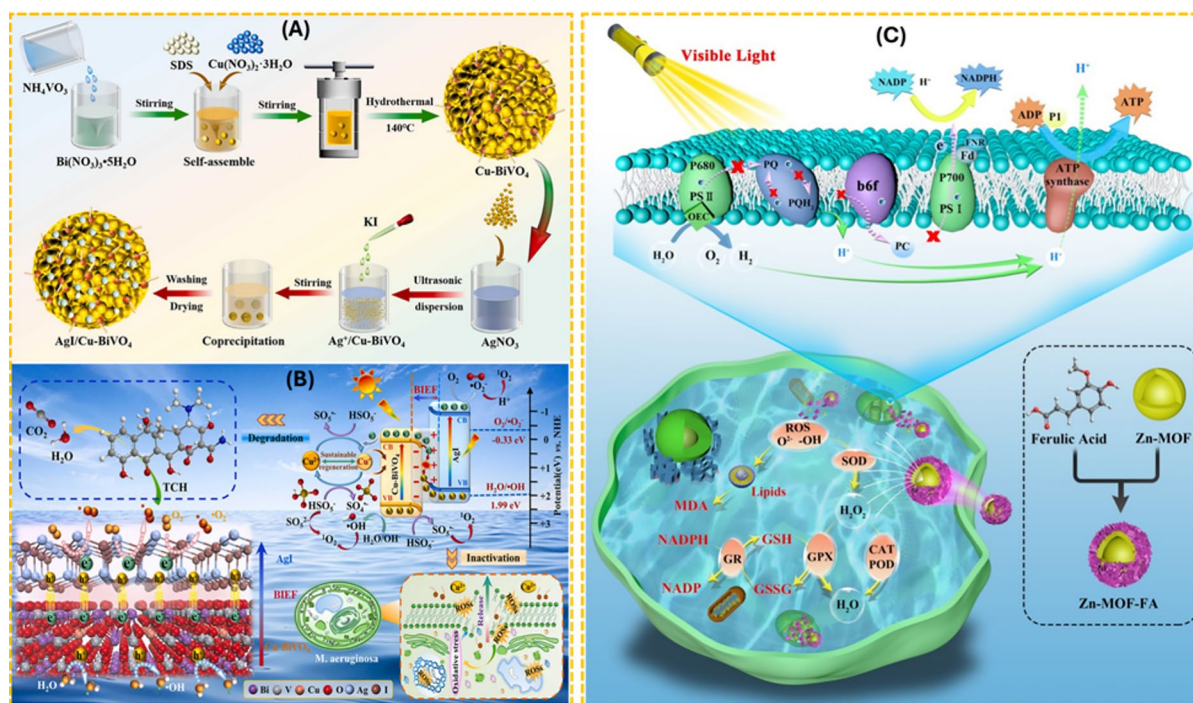


Fig. 6 **A** Representation diagram synthesis of AgI/Cu-BiVO₄; **B** photocatalytic mechanism diagram of tetracycline removal and *M. aeruginosa* inactivation by AgI/Cu-BiVO₄ (reprinted with permission from Li et al. (2024)); and **C** schematic rep-

resentation of photocatalytic mechanism of HAB inactivation via Zn-MOF-FA (reprinted with permission from Wang et al. (2021b))

3.5 AgI/Cu-BiVO₄ S-scheme heterojunction

AgI/Cu-BiVO₄ S-scheme heterojunction catalysts enriched with oxygen vacancies were established by Li et al. (2024) to advance the photocatalytic removal of organic pollutant and *M. aeruginosa* inactivation with PMS support (Fig. 6A, B).

The doping of Cu and AgI coupling in BiVO₄ enabled explicit interfacial charge transmission and plentiful active cores. The catalyst attained 97.6% and 78.8% removal of tetracycline under PMS-supported visible light and algal elimination within 20 min, respectively. The S-scheme assembly permitted effective carrier transmission and robust oxygen reduction, whereas charge separation and activation of PMS were boosted by the Cu²⁺/Cu⁺ redox cycling and surface oxygen defects. These aspects improved mass transference and catalytic efficacy during non-homogeneous systems. This work offers an understanding of designing efficient S-scheme photocatalysts and widens their environmental applications.

3.6 Metal organic frameworks-based structures

Metal–organic frameworks (MOFs) are a well-recognized functional material. MOFs have established their substantial efficiency for adsorption, catalysis, drug delivery, sensing, energy, antimicrobial, and supercapacitor applications (Fonseca and Gong 2022; Chu et al. 2025). For HAB inhibition, MOFs hold numerous superior characteristics, viz., easy to synthesize, functionalize, and modify, can be activated for simultaneous generation of holes (h⁺) and ROS, the optimal growth temperature of HAB results in the increase of MOF's catalytic activity, and exhibits multiple inhibition mechanisms toward HAB (Chu et al. 2025). MOF-composites for *M. aeruginosa* are categorized further into three groups.

3.6.1 Pristine MOFs

Numerous metal–organic frameworks (MOFs) are employed to prevent *M. aeruginosa* growth (Fan et al. 2018). Cu-MOF-74 displayed the maximum algae

inhibition. Studies showed that the Cu^{2+} slow release activated the intracellular generation of ROS, which was recognized as a main repressive factor. Furthermore, when MOFs attain a nanoscale size, they may enter algal cells directly. Li and Wang (2024) established this by detecting the presence of UiO-66- NH_2 particles (40–380 nm) inside *M. aeruginosa* cells. The organic matter release was facilitated by the presence of elevated concentrations (>20 mg/L) of UiO-66- NH_2 intracellularly. However, injured algal cells might discharge microcystins into the water, raising the threat of possible secondary pollution. Moreover, Cu^{2+} , Zn^{2+} , and Ag^+ ions can non-selectively disturb other microbes, whereas lethal metals such as Cd^{2+} cause critical environmental jeopardies. Therefore, the vigilant choice of metal salts and specific control over the particle size of MOF during production is vital. The elimination of *M. aeruginosa* in the presence of light irradiation has also been carried out by pristine MOFs. Yue et al. (2024) employed Cu-MOF-1-4 photocatalysts of variable particle dimensions via regulating reaction conditions. Fascinatingly, their results opposed common drifts, as small particle dimensions of MOF did not essentially improve the algal elimination efficacy.

3.6.2 MOFs-based composites

Even though pristine MOF catalysts have shown promising potential for HAB removal, they still suffer from inadequate light absorption and the quick recombination of h^+ and e^- in photocatalytic processes, they have shown a promising potential for HAB removal (Chu et al. 2025). To address these issues, heterojunction and composite fabrication are considered as required approaches. The photocatalytic is an emergent and effective inactivation technique to control *K. mikimotoi*. Bi_2O_3 @copper MOF (Cu-MOF) composite was synthesized by Wang et al. (2023a) to inactivate *K. mikimotoi* in the visible light. The combination of Cu-MOF decreased the band gap, increasing the adsorption of visible light. Bi_2O_3 @Cu-MOF-1 (60 mg/L), having a specific surface area and a mesoporous pore size of $276.604\text{ m}^2/\text{g}$ and 4 nm, respectively, attained 96.35% removal of *K. mikimotoi* in 4 h. The catalyst also decreased chlorophyll content and soluble protein and disrupted the antioxidant system. Wang et al. (2021b) reported the repressing effect and mechanism of *M.*

aeruginosa inactivation by uniting a MOF with the ferulic acid (FA) (allelochemical). The outcomes validated that the collaborative interaction amid Zn^{2+} and FA in Zn-MOF-FA improved the *M. aeruginosa* inactivation more efficiently compared to Zn^{2+} or FA actions alone. *M. aeruginosa* inactivation exceeds 50% after four days with 2 mg/L of Zn-MOF-FA concentration. The main mechanisms comprise interruption of the PS, together with oxidative damage caused via ROS (Fig. 6C).

3.6.3 MOFs-based floating monoliths

Copper sulfate (CuSO_4) is extensively utilized to monitor HABs owing to its fast and efficient algicidal activity. However, its implication can inadvertently damage non-target aquatic microbes, emphasizing the necessity for innocuous replacements. Kim et al. (2023a, b) synthesized Cu-based MOF chips from waste polyethylene terephthalate bottles. The chips efficiently suppressed the growth of *M. aeruginosa*, signifying a similar activity to the pristine CuSO_4 algicide. Furthermore, these MOF chips do not display any acute toxicity towards *Daphnia magna*. ROS generation by Cu-BDC MOF and Cu_2O causes aggregation of cyanobacteria, disturbance in photosynthesis, and following growth suppression. Cu-BDC MOF chip is an eco-friendly and sustainable method to alleviate HABs without causing substantial harm to aquatic ecosystems. A floating photocatalyst ($\text{MS}/\text{Bi}_2\text{O}_3$ @Cu-MOF) was developed by Wang et al. (2024) by immobilizing a Bi_2O_3 @Cu-MOF onto a melamine sponge utilizing sodium alginate. To inactivate *M. aeruginosa* in the presence of visible light. At an optimum loading capacity, $\text{MS}/\text{Bi}_2\text{O}_3$ @Cu-MOF- $\text{CA}_{0.4}$ (0.4 g) displayed robust photocatalytic activity, attaining a 74.4% *M. aeruginosa* removal rate after 120 h. The generation of hydroxyl ($\cdot\text{OH}$) and superoxide ($\cdot\text{O}_2^-$) radicals played a vital part in the destruction and inactivation of the algal cells. Table 4 represents the utilization of different MOF-based materials for algae removal.

3.7 Graphite/graphene-based materials

Cheng et al. (2024) developed a chitosan-graphene (CG) matrix laden with Fe_3O_4 nanoparticles (MCG) and immobilized microcystinase A (MlrA@MCG) to eliminate detrimental microcystins in water. MlrA@MCG established greater removal efficacy, improved resistance towards heat and alkaline, and

Table 4 Different MOFs-based materials for algae removal

Material type	MOFs-based materials	Target algae	Dominant inhibition mechanism	References
Pristine MOFs	Cu-MOF-1, Cu-MOF-2, Cu-MOF-3, Cu-MOF-4	<i>Microcystis aeruginosa</i>	Photocatalytic ROSs oxidation, hetero-aggregation, metal ion release	Yue et al. (2024)
Pristine MOFs	UiO-66-NH ₂	<i>Microcystis aeruginosa</i>	Nanotoxicity by cell internalization	Li and Wang (2024)
Pristine MOFs	Cu-MOF-74	<i>Microcystis aeruginosa</i>	Metal ion release, hetero-aggregation, ROSs generation	Fan et al. (2019a)
Pristine MOFs	MIL-101(Cr)-NH ₂	<i>Microcystis aeruginosa</i>	Hetero-aggregation	Li et al., (2022a, b)
Pristine MOFs	MIL-101(Cr)-NH ₂	<i>Chlamydomonas reinhardtii</i>	Hetero-aggregation	Li and Wang (2023)
Functionalized MOFs	Zn-MOF-FA	<i>Microcystis aeruginosa</i>	Allelopathic effect, metal ion release	Wang et al., (2021b)
MOFs-based heterojunctions	Ag/AgCl@ZIF-8	<i>Microcystis aeruginosa</i>	Photocatalytic ROSs oxidation	Fan et al. (2020b)
MOFs-based heterojunctions	Bi ₂ O ₃ @Cu-MOF	<i>Karenia mikimotoi</i>	Photocatalytic ROSs oxidation	Wang et al., (2023a)
MOFs-based heterojunctions	NZVI@PCN-224	<i>Pseudomonas aeruginosa</i> , <i>Chlorella vulgaris</i>	Photocatalytic ROSs oxidation	Zhao et al. (2023)
MOFs-based heterojunctions	AgBr/NH ₂ -MIL-125(Ti)	<i>Microcystis aeruginosa</i>	ROSs generation	Cao et al. (2024)
MOF-based composite	CeMOFs	-	Adsorption	Chen et al. (2022)
MOFs-based composites	SNP-TiO ₂ @Cu-MOF	<i>Karenia mikimotoi</i>	Photocatalytic ROSs oxidation, hetero-aggregation	Hu et al. (2023)
MOFs-based monoliths	Ag/AgCl@ZIF-8 floating coating	<i>Microcystis aeruginosa</i>	Photocatalytic ROSs oxidation	Fan et al. (2019b)
MOFs-based monoliths	MS/Bi ₂ O ₃ @Cu-MOF	<i>Microcystis aeruginosa</i>	Photocatalytic ROSs oxidation, adsorption, metal ion release	Wang et al. (2024)
MOFs-based monoliths	Co-MOF/PVA polymeric network	<i>Navicula</i>	Unknown	Shahryari et al. (2022)
MOFs-based monoliths	Cu-BDC MOF chips	<i>Microcystis aeruginosa</i>	Metal ion release, ROSs generation from Cu ₂ O	Kim et al., (2024b)
MOFs-based monoliths	ZIF-8/chitosan/melamine sponge	<i>Nitzschia</i>	Membrane permeability change by chitosan	Guo et al. (2022)

easy recovery and reusability even after 10 cycles with 48.6% activity. Remarkably, no cytotoxicity was observed towards human cells by MlrA@MCG and its degradation products. In simulated *Microcystis* blooms, MC-LR was decreased from 232.64 µg/L to 94.39 µg/L in 24 h by MlrA@MCG (0.2 U), showing its catalytic constancy. This investigation grants an innocuous and efficient approach to

treating microcystin, presenting a promising answer to improve water quality and ecological security.

Zetterholm et al. (2022) studied chitosan/graphene (CSG) composites to eliminate algae (*Anabaena* sp., *Synechocystis* sp., and *M. aeruginosa*) and toxins. Depending on the species, algal elimination (40–90%) is attained with high microcystin removal efficacy, removing over 94% of MC-LR. Analyses

(SEM and XRD) exposed that graphene improves the chitosan matrix, allowing π - π interactions between graphene and the aromatic ring of microcystin. This mechanism proposes that CSG is an effective and sustainable biopolymer to remove HAB-polluted waters, presenting a promising approach for water cleansing and toxin removal. A graphene film composite biosensor was developed by Zhang et al. (2018a) for fast, precision detection of MC-LR. It established outstanding linearity ($R^2=0.99$) through 0.005–10 $\mu\text{g/L}$, including a 2.3 ng/L detection limit. Extreme surface area and conductivity allowed great performance, and effective confirmation in environmental waters emphasizes its scalability and real monitoring potential. Roberts et al. (2025) exhibited that mesoporous graphene nano-platelets (GnPs) effectively removed the saxitoxin, with a 93.5-fold better adsorption capacity and sixfold quicker kinetics compared to the granular activated carbon. The high adsorption capacity is due to the high surface area, optimized porosity, and π - π /electrostatic interactions.

3.8 g- C_3N_4 -based materials

In the field of environmental catalysis, CN is recognized among the most valued nanomaterials owing to its astonishing ascendancy (Feng et al. 2022, 2024). Zhang et al. (2023a, b) developed an innovative approach that combines ZnO with C_3N_4 composites for the effective collection of intact *M. aeruginosa* cells. The effect of different concentrations of g- C_3N_4 precursors was evaluated. Approximately 93% of algae cells were removed by the ZnO/ C_3N_4 (2:1) system within 10 min at a dose of 4 g/L. Further, expanded perlite (EP) was used as a floating carrier to create EP@n- $\text{C}_3\text{N}_5/\text{Ag}_2\text{CO}_3$ (EP@CNAC), a unique spherical floatable Z-scheme core-shell photocatalyst (Li et al. 2025). EP@CNAC-3 was capable of destroying 97% of microcystin-LR in 16 min owing to defect engineering and Z-scheme heterojunctions, which greatly improved carrier migration kinetics and solar spectrum reactivity (Fig. 7). DFT studies showed a distribution of electron density and carbon defect sites

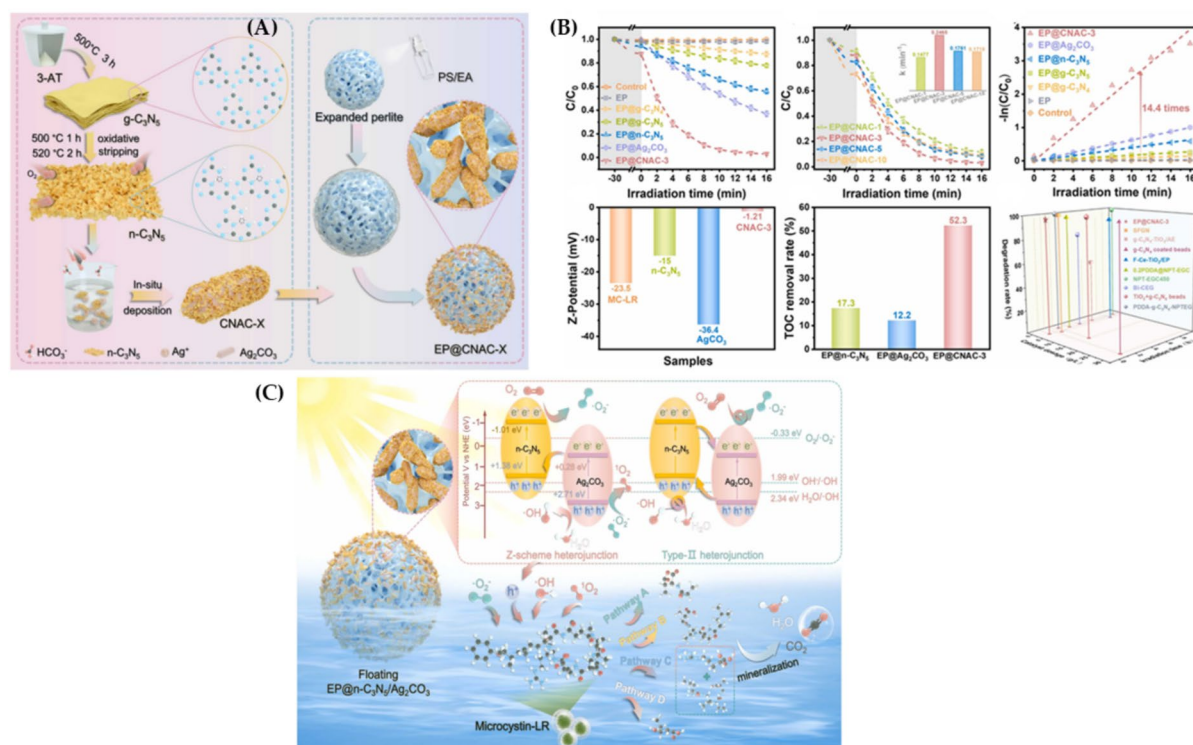


Fig. 7 A Synthesis of floating EP@CNAC; B photocatalytic degradation efficiency of MC-LR, and C photocatalytic mechanism of MC-LR removal (reprinted with permission from Li et al. (2025))

in C_3N_5 . Mechanistic investigations using LC–MS, Fukui index, and electrostatic potential examination demonstrated that 1O_2 electrophilically cleaved the Adda side chain, while h^+ and $\bullet O_2^-$ caused nucleophilic ring-opening and decarboxylation.

$Ag_2O/g-C_3N_4$ photocatalysts with p-n heterojunctions were produced to eliminate algae in the presence of visible light (Fan et al. 2020b). They assessed the influences of surface charge and configuration ratio on the development of heterojunction and illustrated its optical properties, crystal phase, morphology, functional groups, and elemental composition. $Ag_2O/B-g-C_3N_4$ demonstrated the utmost photocatalytic activity (14.3 wt%), efficiently eliminating *M. aeruginosa*. $g-C_3N_4$ instantly formed complex with Ag_2O , and the Ag_2O to $g-C_3N_4$ ideal composition was 1:6, accomplishing 99.94% *M. aeruginosa* elimination within 6 h. Superoxide dismutase and malondialdehyde concentrations were reduced, but an increase in catalase concentration was observed, indicating the downfall of the antioxidant system. The broken cell membrane of algae discharged K^+ , Ca^{2+} , Mg^{2+} ions, lowering the concentration of chlorophyll a and phycoobiliprotein and harming the photosynthetic system.

Fan et al. (2021c) prepared $Ag/AgCl@g-C_3N_4@UIO-66(NH_2)$ heterojunction for the photocatalytic inactivation of HABs. The nanocomposite provided too many active sites along with improved dimensions to use visible-light and efficiently transfer charge carriers. This resulted in enhanced removal efficiency of cyanobacteria (99.9%) degrading within 3 h. Further, the cycle test showed exceptional reusability and stability. A $Pd/g-C_3N_4$ nanocomposite was developed by Lu et al. (2022a) assisted fast, visible-light-free elimination of *M. aeruginosa*, showing 95.17% removal of cells within 10 min. Membrane disturbance was proven during potassium release, microcystin leakage, and TEM analysis. Later, MC-LR elimination further displayed effectiveness. This Pd-driven coagulation–breakage mechanism extends a favorable, quick, and realistic approach for cyanobloom control. Zhang et al. (2025a) synthesized a $g-C_3N_4/calcite/pumice$ floating photocatalyst for the elimination of *M. aeruginosa* under visible light. Results showed 97.56% inactivation of *M. aeruginosa* due to the generation of $\bullet O_2^-$ and $\bullet OH$ radicals. The radicals interrupted membranes, prevented antioxidant enzymes, and destroyed the MC-LR. The composite also confirmed strong reusability and low cytotoxicity.

A novel magnetically divisible $ZnFe_2O_4/Ag_3PO_4/g-C_3N_4$ with a double Z-scheme was synthesized to inactivate *M. aeruginosa* and MC-LR in visible light (Fan et al. 2022b). ZFO/AP/CN (100 mg/L) degraded 96.33% of chlorophyll after 3 h of visible light irradiation. Malondialdehyde concentration was intensified during the process, whereas superoxide dismutase and catalase levels primarily ascended and later decreased. The catalyst effectively removed *M. aeruginosa* (94.31%) and MC-LR (76.92%). Scavenging experiments verified that $\bullet O_2^-$ and $\bullet OH$ were essential for algae removal and MC-LR degradation (Fig. 8). The system has proven tremendous recoverability and steadiness via cycling tests employing magnetic revival. In general, ZFO/AP/CN demonstrated incredible photocatalytic functioning in visible light and delivered a substantial capability for removing HABs.

3.9 Zeolite-based adsorbent

Zeolite is widely accessible and effective in immobilizing NPs, decreasing secondary pollution (Azambre et al. 2020). Owing to the presence of a negative charge on via zeolite, the immobilization of NPs possessing the positive charge, viz. Ag^+ ions provide stability to the adsorbent. Zeolite's adsorption characteristics mostly depend on its particular chemical and structural composition. The adsorption process is significantly controlled via Si/Al ratio, cation type, number, and position. Fernandes et al. (2024) investigated the adsorption of *M. aeruginosa* cells from an aqueous system using modified homoionic zeolites impregnated with silver nanoparticles (ZH-AgNPs).

3.9.1 Covalent organic frameworks-based structures

In comparison to the conventional photocatalysts (viz. TiO_2), MOFs and COFs-based photocatalysts are able to use visible light. Higher SSAs and appropriate pore structures tend to have more active sites for photocatalysis, and further research for developing new potential and durable composite materials. Here, for HAB inhibition, COFs are still less researched than MOFs. As most of the MOFs primarily focused on the red tide HAB inhibition (*K. mikimotoi*) and water wash toxic HABs (*M. aeruginosa*), COFs are researched principally about the inhibition of phyco-toxins and microcystins. Salonen et al. (2017) used TpBD-Me₂ COF for absorbing the marine toxin

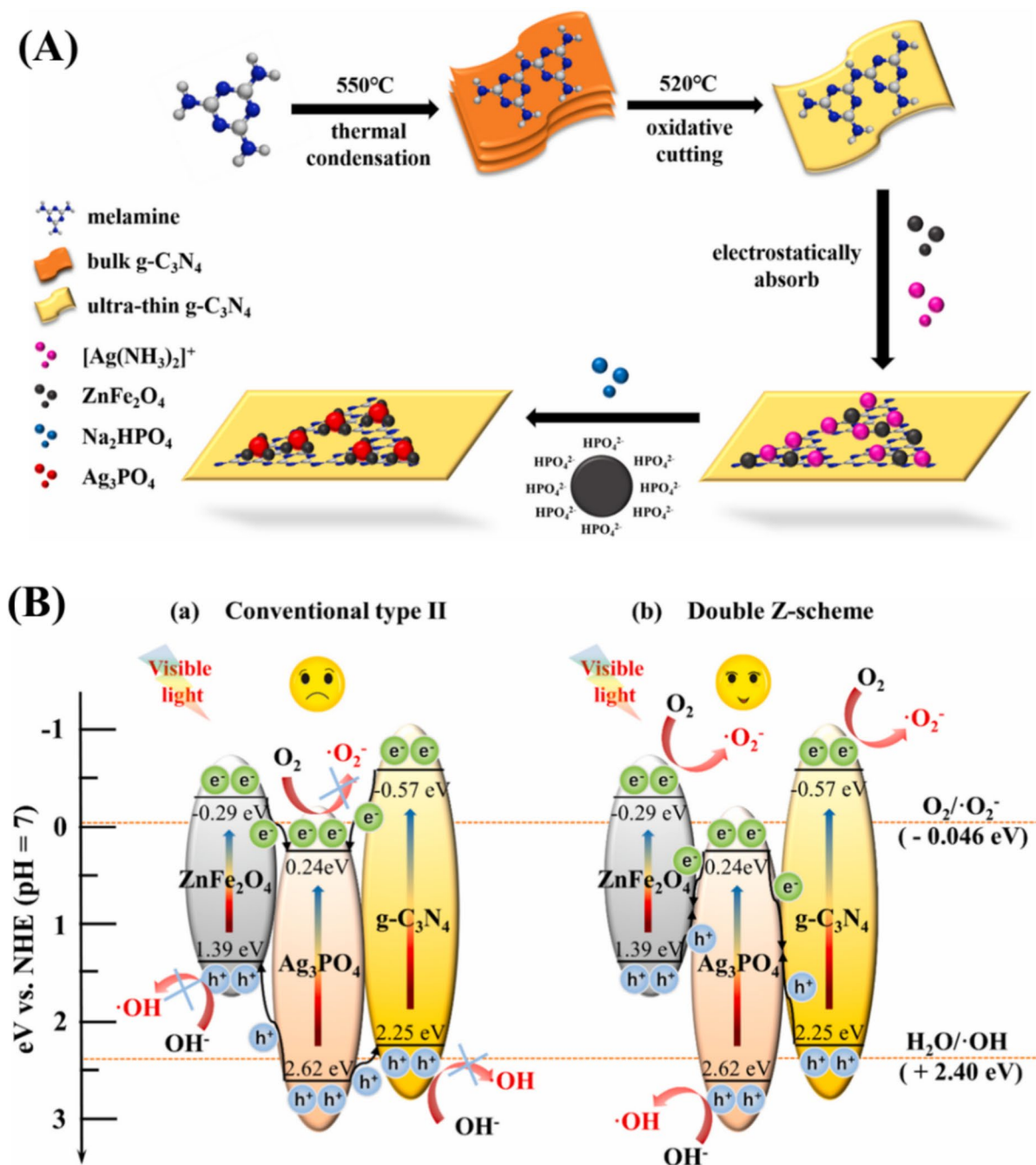


Fig. 8 **A** Representation diagram synthesis of ZFO/AP/CN; **B** photocatalytic mechanisms for inactivation of *M. aeruginosa* (reprinted with permission from Fan et al. (2022b))

okadaic acid (OA). Here, within 1 h, equilibrium concentration was attained, reaching 61 mg/g; the highest adsorption capacity. Furthermore, both ethanol (70%) and acetonitrile dissolved OA, and the COF materials

were recycled for reusability. Presently, many studies are still under trial using COFs to determine the whole aspect of photocatalytic HABs toxin inhibition.

Table 5 Current findings on biochar for nutrient adsorption in eutrophic waters influenced by algal blooms

Biochar Type	Synthesized Method	Nutrient Type	Adsorption capacity (mg g ⁻¹)	References
Rice straw	Carbonization	NH ₄ ⁺ -N	13.21–13.91	Gong et al. (2019)
		NO ₃ ⁻ -N	10.38–13.40	
		P	12.18–12.42	
		K	11.51–13.50	
Oak wood, press cake	Hydrothermal carbonization	PO ₄ -P	0–30	Takaya et al. (2016)
		NH ₄ ⁺ -N	105.8–146.4	
Chipped hemp fiber	Pyrolysis	PO ₄ -P	ND	Bolton et al. (2019)
Pine-derived biochar	–	Microcystin-LR and saxitoxin	> 90%	Chambers et al. (2024)
Pine-derived biochar	Pyrolysis	Saxitoxin	> 90% (q _m of 314.37 µg/g)	Chambers et al. (2025)
Polyacrylamide-grafted gelatin/biochar/Fe ₃ O ₄	–	Microcystin-LR	88%	Yao et al. (2025)
Rice straw derived biochar	Pyrolysis	Microcystin-LR	10.96 µg/g	Wei and Lu (2021)

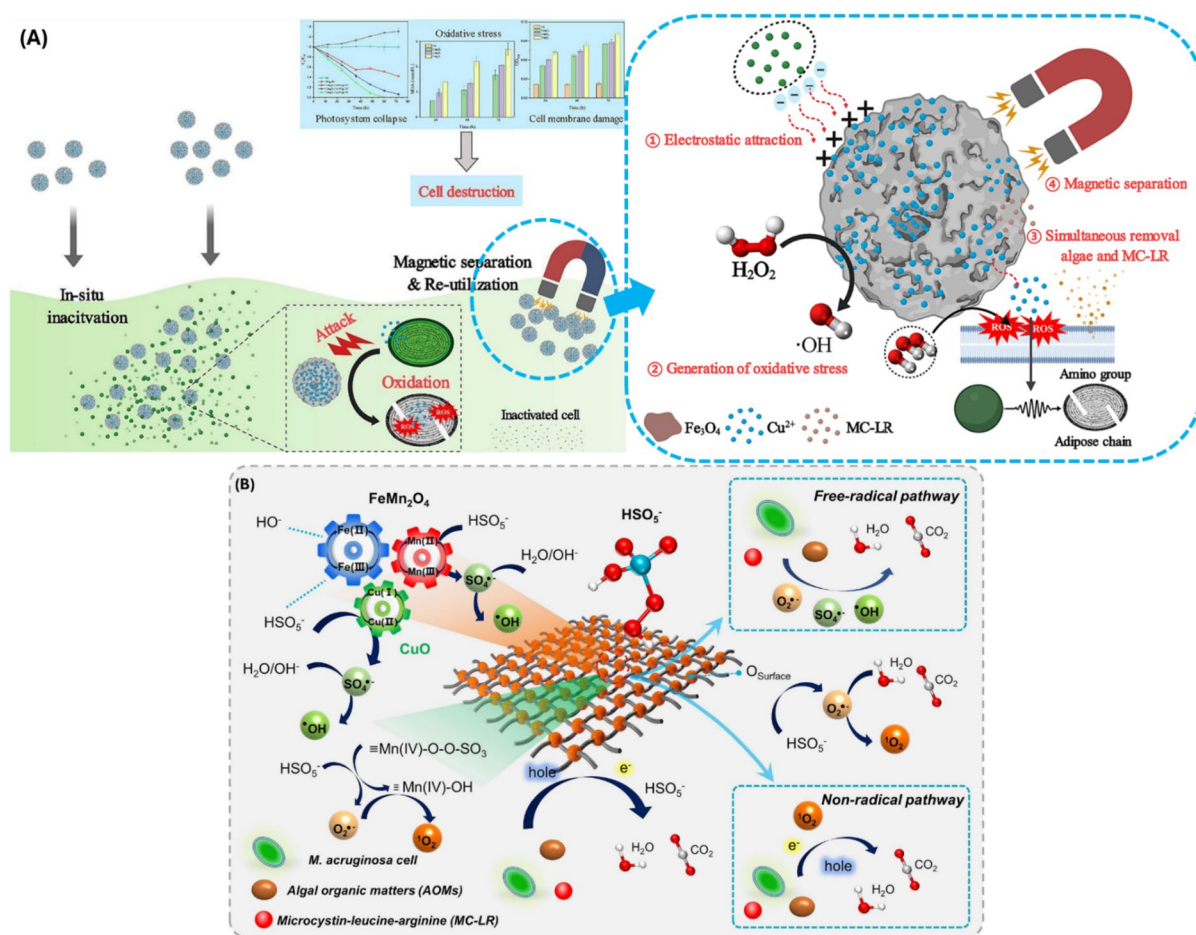


Fig. 9 **A** Recyclable Cu²⁺ doped Fe₃O₄@biochar for in-situ inactivation of *M. aeruginosa* (reprinted with permission from Gao et al., (2024a). **B** Mesoporous carbonaceous framework

supported Fe-Cu-Mn oxides for the control of HABs (reprinted with permission from Yang et al. (2023c)

3.9.2 Biochar

Liu et al. (2023) explored $\text{Fe}_3\text{O}_4\text{-BC@Cu-MOF-74}$, which is a biochar-modified copper MOF doped with Fe_3O_4 to inhibit *M. aeruginosa*. This material was slender, permeable, and firm, mainly comprising C, O, Fe, and Cu, with separate crystal planes and having C–O, Cu–O, O–H functional bonds. $\text{Fe}_3\text{O}_4\text{-BC@Cu-MOF-74}$ attained a 32.2% removal efficacy and decreased chlorophyll-a content by 60% in the visible light. The oxygen and hydroxyl radicals were produced, initiating oxidative stress and enhancing the antioxidant activity, causing elevated soluble protein. Additionally, cell membranes were disturbed by Cu^{2+} , Fe^{2+} , and Fe^{3+} , instigating break and death. This study emphasizes $\text{Fe}_3\text{O}_4\text{-BC@Cu-MOF-74}$ as a promising explanation for monitoring *M. aeruginosa* below visible light, delivering a substitute to prevailing algae elimination approaches that are expensive and unsuccessful. Wei et al. (2023) assessed multifunctional biochars, i.e., magnesium-biochar (Mg-biochar), rhamnolipid-biochar (RL-biochar), and Mg-RL-biochar to effectively remove phosphate, petroleum hydrocarbons, and harmful algae. Mg-RL-biochar displayed the maximum algal inhibition (61–64% biomass reduction), compelling it an extremely favorable combined treatment material. Table 5 represents some recent studies on biochar for nutrient adsorption by algal blooms.

Cu^{2+} -doped $\text{Fe}_3\text{O}_4\text{@biochar}$ magnetic composite (Cu-Mag-BC) was reported by Gao et al. (2024a) for deactivation of *M. aeruginosa*. Cu-Mag-BC quickly deactivated chlorophyll-a at 5 mg/L of dosage with slight leaching of Fe and Cu. ROS were produced mainly $\cdot\text{O}_2^-$ and $\cdot\text{OH}$ causing oxidative stress and fluctuating SOD and catalase activities. Over three cycles ~80% inactivation was maintained by Cu-Mag-BC, accomplishing a 2.84 g/g total chl-a elimination efficacy (Fig. 9A). The suggested mechanism proposes that amino groups and adipose chains as the main oxidation locations. BCs derived from corn stalks were doped with Cu^{2+} to produce Cu-BC composites, used as an algicide by Gao et al. (2024b). *M. aeruginosa* growth was inhibited by biochar produced at elevated temperatures ($\text{BC550} \gg \text{BC750} > \text{BC950}$), whereas low temperature (BC350) produced biochar encouraged growth as a result of nutrient release. The suppression of growth was in relation to physicochemical characteristics, including nutrient

adsorption, surface area, and tenacious free radicals. The composite displayed improved deactivation, but efficacy fluctuates with Cu^{2+} load and pyrolysis temperature. This work established that both pristine BCs and Cu-BCs efficiently remove algae, principally affected via pyrolysis temperature. Wei et al. (2025) developed a *Chlorella*-derived biochar-supported nZVI (BC/nZVI) composite to remove *M. aeruginosa*. The composite attained achieving 92.27% removal of *M. aeruginosa* due to intracellular Fenton-driven ROS generation ($^1\text{O}_2$, $\text{O}_2^{\cdot-}$, $\cdot\text{OH}$), which impaired antioxidant enzymes, encouraged lipid peroxidation, and initiated membrane rupture and metabolic collapse.

Due to its unique qualities, biochar has a lot of promise to mitigate algal blooms. It can absorb extra phosphates and nitrates from eutrophic water bodies affected by HABs owing to its enormous surface area (Novais et al. 2018). Furthermore, the porous nature of biochar encourages the colonization of microbial populations, which are essential for decomposing organic matter and generating compounds that inhibit the growth of algae. The ability of biochar to prevent algal blooms is further strengthened by this microbial interaction (Cheng et al. 2023). The capacity of biochar to absorb and immobilize nutrients, particularly N and P, that are critical for algae growth is a major factor in its efficacy. Owing to its enormous surface area and porosity, biochar limits the growth of algae by lowering the accessibility of nutrients in the water column. Crucially, when the temperature at which biochar is pyrolyzed rises, its capacity to absorb nutrients improves as well, making it more effective at eliminating phosphate and ammonium ions from eutrophic waters (Ren et al. 2021; Kończak and Huber 2022).

Additionally, biochar speeds up the sedimentation process of algae by promoting their aggregation. Flocculation results from electrostatic interactions between the surfaces of biochar particles and algal cells. By creating bigger aggregates that settle faster, this mechanism shortens the duration that algae spend in the water column and lessens their effects (Gao et al. 2024b). Biochar also alters the chemistry of the water and competes with algae for nutrients. Biochar helps prevent blooms by reducing the growth of some algae species by changing the pH of the water and releasing organic molecules (Zhu et al. 2019). Biochar influences microbial ecology in aquatic environments in addition to nutrient absorption

and aggregation. HAB growth is indirectly controlled via microorganisms that benefit from biochar amendments.

Certain bacteria may directly interact with algal cells, changing their growth dynamics and community structure, or they may outcompete algae for resources. In these interactions, biochar promotes two primary mechanisms: direct impacts via physical contact with algae and indirect effects through microbial activity. In order to provide nutrients or improve nutrient cycling, for example, some bacteria may form mutualistic partnerships with algae, which can occasionally promote algal development. On the other hand, by generating inhibitory chemicals or competing for nutrients, other bacteria can stop the growth of algae (Cheng et al. 2022; Liu et al. 2023).

Cheng et al. (2022) demonstrated how well biochar works in built wetlands (CWs) to reduce eutrophication and eliminate microcystins. To assess the effects of biochar, the investigators employed cylindrical polyethylene microcosms with different degrees of biochar addition: 0%, 10%, 20%, and 50%. Biochar-amended wetlands improved nutrient removal ($\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$) and caused changes in the composition of the microbial community, according to two three-day batch studies. Functional microorganisms that are known to suppress algal development and lessen water eutrophication, including *Burkholderiaceae*, *Micrococcaceae*, *Nitrospiraceae*, *Xanthomonadaceae*, and *Sphingomonadaceae*, were found to be more prevalent in the study. The highest concentration of these advantageous microorganisms was responsible for the best performance of the BC20-CWs and BC50-CWs. These results highlight the dual function of biochar in effectively reducing eutrophication and algal blooms by interacting directly with algae and cultivating microbial communities.

4 Others composites

4.1 Polyvinyl chloride composites

An adsorption-based method was employed to alleviate HABs without instigating secondary contamination (Park et al., 2022b). They developed polyethylenimine–polyvinyl chloride (PEI–PVC) composite fiber, to eliminate 90% and 80% cyanobacterium efficiently (*M. aeruginosa*) with a negatively charged

surface in artificial media and in real water, respectively, without cell lysis. Reduction in microcystin concentration was also observed through concurrent adsorption. The PEI–PVC was regenerated via alkaline desorption, preserving its elimination efficacy. Unlike pristine PVC, no toxic chlorine was released from the composite fiber under light exposure. This adsorption-based method offers an even more effective technique for monitoring HAB pollution in water resources.

4.1.1 Metals-mesoporous carbon framework composites

Few researchers have explored advanced oxidation processes (AOPs) to monitor HABs utilizing heterogeneous catalysis (Yang et al. 2023a, b, c). Yang et al. (2023c) synthesized a mesoporous carbon framework supported Fe–Cu–Mn oxides for peroxy-monosulfate (PMS) activation for eradicating *M. aeruginosa* and its organic matter. More than 99% of *M. aeruginosa* removal at low FCMT and PMS dosages in the FCMT/PMS within 2 h and pH (3–9). Even though the FCMT/PMS approach encouraged oxidative stress, harming cell membranes and successfully eliminating DOC (over 50%) and MC (54%), thus blocking serious secondary contamination from cell lysis (Fig. 9B). The removal of algae is carried by both free and non-free radical paths, the principal ROS observed was $\bullet\text{O}_2^-$ and $^1\text{O}_2$.

4.1.2 TiO_2 -biochar composites

Innovative elimination approaches are required to circumvent unnecessary algacide usage and ecological jeopardies. A hybrid method was employed by Wang et al. (2021a) compounding TiO_2 /biochar (TiO_2 /BC) with ultrasound irradiation to efficiently remove *M. aeruginosa* after coagulation. This hybrid process eliminated 92% of algal cells while averting a substantial surge in DOC and microcystins. Reactive species augmented cell membrane penetrability, enhancing coagulation competence. The study emphasizes the TiO_2 /BC as a proficient catalyst to control cyanobacteria, presenting an effectual and eco-friendly solution to treat polluted water.

Table 6 A comparison of mechanistic framework for photocatalytic nanomaterials for HAB removal from wastewater (Baird, 2023; He et al. 2021; Song et al. 2023; Wang et al. 2026)

Nanomaterials	Bandgap (eV)	Primary ROS species	HAB damage pathway	Wastewater matrix effects
TiO ₂ nets/fibers	3.0–3.2	-OH, O ₂ ⁻	Membrane lipid peroxidation; cell lysis; chlorophyll a degradation; PS II damage	DOC scavenging of -OH
g-C ₃ N ₄	2.7	¹ O ₂ , O ₂ ⁻ , -OH	Chlorophyll inhibition; lipid oxidation; inhibition of antioxidant activity	Better DOM tolerance
Magnetic composites	2.5–2.8	-OH, O ₂ ⁻ , H ₂ O ₂	Cell wall breaking; DNA impairment	Fe leaching risk
MOFs	2.8–3.2	¹ O ₂ , O ₂ ⁻ , -OH	Pore-confined ROS outbreak; adsorption and degradation of toxin; enzyme disruption	Pore blocking by organics; Cu-MOFs leach in wastewater
COFs	2.4–2.9	¹ O ₂ , O ₂ ⁻ , -OH	Attack of singlet O ₂ on unsaturated lipids/chlorophyll	High stability; tunable pores trap of HAB-LR
MXenes	~ 1.5–2.5	O ₂ ⁻ , -OH	Generation of plasmonic ROS; membrane rupture; metabolic activity degradation	Restacking/oxidation in water; DOM quenches plasmon
Graphene composites	2.5–2.9	O ₂ ⁻ , -OH	π - π adsorption of cells; ROS generation	Graphene fouling via proteins; excellent MC-LR adsorption
Biochar composite (BC/TiO ₂)	2.7–3.0	O ₂ ⁻ , -OH	Adsorption traps cells; localized ROS generation	Excellent DOM tolerance; biochar buffers scavengers

Table 7 Comparative analysis of different photocatalysts on the basis of synthesis cost, scalability, and long-term stability (Serrà et al. 2021; He et al. 2021; Yang et al. 2023a, b, c; Wang et al. 2023b)

Photocatalyst	Technique involved	Cost (per kg)	Scalability	Long-term stability	References
MOFs	Solvothermal	Very high to High	Lab scale	More fragile	Zango et al. (2020); Khan et al. (2023)
COFs	Solvothermal/ionothermal	Very high to High	Mostly Lab scale	Moderately stable to more fragile	Husain et al. (2025); Goswami and Oh (2025)
Graphene (not ultrapure)	Hummers/electrochemical	High to moderate	Emerging scale up	Moderately stable	Bu et al. (2022); Mohammadi et al. (2022)
Biochar	Pyrolysis	Low to very low	Industrial ready	Moderately stable	Goswami and Oh (2025)
Zeolites	Hydrothermal	Low	Industrial ready	Most stable	Hong et al. (2017); Khaleque et al. (2020)
TiO ₂ -based composites	Sol-gel/hydrothermal	Moderate to low	Industrial ready	Most stable	Tiwari et al. (2025)
g-C ₃ N ₄	Thermal polymerization	Low	Emerging scale up	Moderately stable	Husain et al. (2025); Goswami et al. (2025a, b)

4.1.3 Mineral-hydrogel composites

A mineral hydrogel complex with calcium phosphate and wollastonite was developed by Xiao et al. (2022) to adsorb and precipitate phosphorus. A model *Synecoccus elongatus* 2973 was used to study the efficiency of the hydrogel composite. Results showed that the composites reduced dissolved P from 5.1 mg/L to 0.31 mg/L and 75% declined biomass growth. The composite displayed 96% and 91% removal of P when implemented to treat wastewater and runoff,

respectively. Recycled and reutilized as a slow-release manure, the P-enriched composites favored algal growth and lipid accretion without hindering light. This method delivers a sustainable solution to monitor phosphorus contamination.

4.1.4 Z-scheme

Fan et al. (2021b) developed a novel Z-scheme $\text{Ag}_3\text{PO}_4@$ polyaniline (PANI) core-shell photocatalyst to inactivate *M. aeruginosa*. With 10 wt.%

Table 8 A comparison of CAPEX, OPEX, energy demand, catalyst recovery, techno-economic advantage and limitation of different architectures/composites for HAB removal from wastewater (He et al. 2021; Song et al. 2023; Wang et al. 2023c, 2026)

Nanomaterials	Architecture type	Energy demand	Catalyst recovery	CAPEX drivers	OPEX drivers	Scalability	TEA advantages	TEA limitations
TiO ₂ nets/fibers	Immobilization on nets, meshes, loofah fibers	For hydraulics, require solar	Physical skimming	Low-moderate	Low	Side-streams	Low-cost	Fouling in turbid water
g-C ₃ N ₄	Foam-supported sheets; 3D hydrogels	Minimal mixing	Raft retrieval	Moderate	Very low	Open lagoons	Edge over TiO ₂	up-scaling
Magnetic composites (e.g., Fe ₃ O ₄ /g-C ₃ N ₄ , ZnFe ₂ O ₄)	Dispersed NPs with magnetic core	Magnetic separation	Magnetic extraction	Moderate-high	Moderate	Recirculating tanks	Fast recovery	Synthesis/equipment cost
MOFs	Powder or thin films on supports	UV light	Filtration or support retrieval	High	Moderate-high	Lab-to-pilot	High activity	Instability
COFs	Porous films or beads	visible light	Sedimentation/bead retrieval	High	Moderate	Selective toxin removal units	Tunable pores	Up-scaling and cost barriers
MXenes	Freestanding films or hybrids	broad-spectrum light	Membrane filtration	High	High	emerging	Plasmonic boost	Poor stability in water
Graphene composites	Aerogels, fibers	UV/visible	Net retrieval or flotation	Moderate-high	Low-moderate	Flow-through reactors	Conductivity aided charge separation	Cost variability
Biochar composite (BC/TiO ₂)	Beads, floats, or sponge matrices	require solar	Skimming/sedimentation	Low	Very low	Low-cost lagoons	circular economy	Lower activity than synthetics

of PANI in $\text{Ag}_3\text{PO}_4@\text{PANI}$, the photocatalyst displayed the greatest activity for algae elimination with 99.2% chlorophyll *a* degradation in the presence of visible light within 3 h with excellent photostability even after three repetitive times cyclic experimental sets. Also, quenching experiment outcomes showed that the dominant reactive species during the process were $\cdot\text{O}_2^-$ and photoinduced hole (h^+).

Table 6 compares major photocatalysts for HAB removal using a unified mechanistic evaluation framework, and Table 7 for comparing the techno-economic feasibility, scalability, and long-term stability. Due to system-specific cost variability, a semi-quantitative assessment is presented, offering clear comparative insights and identifying the most promising photocatalysts for practical, scalable HAB mitigation. Further, comparing the overall photocatalytic nanomaterials based on to the CAPEX, OPEX, energy demand, catalyst recovery, techno-economic advantage and limitation for HAB removal from wastewater is represented in Table 8.

5 Plausible mechanisms and effect of photocatalysts on the HABs' destruction

5.1 Photocatalytic mechanism of architectonics

The fundamentals of photocatalysis are based on the photoinduced electron–hole pairs. The exposure of a semiconductor photocatalyst to light having a higher energy than its bandgap, VB electrons get excited and move to the CB, causing holes in the VB. Due to this numerous electron–hole pairs are generated (Sun et al. 2023a, b). These charged transferors travel to the surface of the photocatalyst, where dissolved oxygen (O_2) reacts with them, forming $\cdot\text{O}_2^-$, while hydroxyl radicals ($\cdot\text{OH}$) are produced due to interaction amid holes and water molecules (Liu et al. 2022a, b). ROS holds a robust redox ability, permitting them to break down contaminants into smaller molecules or accomplish complete mineralization. But a foremost constraint of individual semiconductors is the high recombination frequency of photogenerated carriers, which decreases photocatalytic competence. The reaction mechanism in VLD photocatalytic method is represented in the equations below.

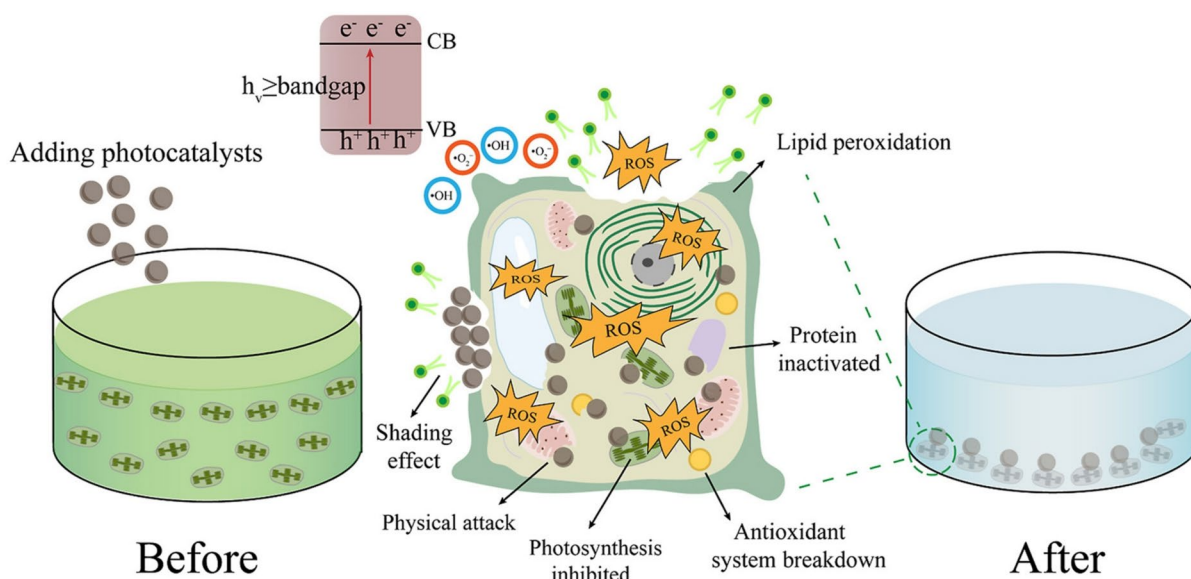


Fig. 10 Schematic representation of the physical limitation effects of photocatalysts and illustration of photocatalysts inactivation of HABs (reprinted with permission from Sun et al. (2023a))

Process	Reaction	Equation number
Generation of e^- - h^+ pairs	$Photocatalyst + h\nu(vis) \rightarrow h^+ + e^-$	(i)
Carrier recombination	$h^+ + e^- \rightarrow Energy$	(ii)
Oxidation	$h^+ + H_2O \rightarrow h^+ + \bullet OH$	(iii)
	$\bullet OH^- + h^+ \rightarrow \bullet OH$	(iv)
Reduction	$e^- + O_2 \rightarrow \bullet O_2^-$	(v)
Formation of H_2O_2	$\bullet O_2^- + H^+ \rightarrow \bullet OOH$	(vi)
	$e^- + \bullet OOH + H^+ \rightarrow H_2O_2$	(vii)
	$\bullet OOH + \bullet OOH \rightarrow H_2O_2 + O_2$	(viii)
Reduction	$e^- + H_2O_2 \rightarrow \bullet OH + OH^-$	(ix)
Formation of 1O_2	$\bullet O_2^- + h^+ \rightarrow O_2^1$	(x)
Algae removal	$ROS(O_2^1/e^-/h^+/\bullet O_2^-/\bullet OH^-/H_2O_2/H^+) + algalcells \rightarrow Degradationproducts$	(xi)

The actions and effects of photocatalysts on freshwater and marine algae have been extensively studied. Biological markers that represent cellular physiological features, viz., growth rate, chlorophyll content, antioxidant enzyme activities, membrane integrity, and the rate at which photocatalytic particles are assimilated, are frequently utilized for assessing the detrimental effects of photocatalysts on algae (Morelli et al. 2018; Sendra et al. 2018). According to earlier research, photocatalysts work via two main mechanisms to prevent algal growth or even cause cell death: physical constraints (Hu et al. 2018; Sendra et al. 2017a) and oxidative stress brought on by ROS (Middepogu et al. 2018; Xiao et al. 2022). Despite these, physical effects are less important in algal cell inactivation than oxidative stress, which mostly happens during photocatalytic activities (Li et al. 2015). In contrast, physical constraints can be broadly classified into the following: direct damage to membranes and cellular structures (Xia et al. 2015), sedimentation effects, aggregation phenomena (homoagglomeration and heteroagglomeration) (Sendra et al. 2017a; Wang et al. 2016), and shading effects.

5.2 Effect of photocatalysts on algae cells

5.2.1 Oxidative stress and damage

ROS, which can damage algal cells, are largely produced by photocatalytic processes. When exposed to light, photocatalysts produce ROS, which are plausibly the prime cause of harming the HAB. Light or photons with energy equivalent to or more than

the semiconductor bandgap activate semiconductor catalysts during the photocatalytic process (Banerjee et al. 2014; Nakata and Fujishima 2012). Electron-hole (e^- - h^+) pairs are created as a result of this excitation and go to the photocatalytic particles' surface. Interfacial interactions take place at the surface, such as oxidative reactions between h^+ and surface-adsorbed water or hydroxide ions and reductive reactions among e^- and surface-adsorbed oxygen (Guo et al. 2019). Intense ROS are produced by these processes, including hydrogen peroxide (H_2O_2), hydroxyl radicals ($OH^\bullet$), hydroperoxyl radicals ($HOO^\bullet$), superoxide radicals ($\bullet O_2^-$), and singlet oxygen (1O_2) (Dong et al. 2015).

Several heterogeneous photocatalysts have been produced to prevent harmful cyanobacteria from proliferating. The photocatalytic ability of nano-TiO₂ in photokilling *M. aeruginosa* was examined by Chen et al. (2018). According to their research, algal cells' metabolic activity dramatically declined with longer photocatalysis times, and after six hours, there was noticeable cell disintegration. Serrà et al. (2020) created the hybrid photocatalyst Ni@ZnO@ZnS and evaluated how well it worked against *Anabaena flos-aquae* and *Spirulina platensis paracas* in the presence of sunshine. Excellent antibacterial activity was demonstrated by this hybrid photocatalyst. A catalyst dosage of 0.6 mg/mL eliminated almost all algal cells in 50 min of photocatalysis when the initial concentration of microalgae was $100 \pm 10 \times 10^3$ cells/mL. In order to assess the photocatalyst's use in natural settings, additional studies examined microalgal biomass. Over the course of two days, biomass in the

photocatalytic system decreased somewhat in comparison to controls. The photocatalytic capabilities of MOFs are also investigated. Fan et al. (2019a) created Cu-MOF-74 and investigated how it inhibits dangerous cyanobacteria. In *M. aeruginosa*, Cu-MOF-74 promoted growth at low doses. But at dosages greater than 1 mg/L, relative extracellular ROS levels rose up to 435% in comparison to controls, and the inhibition rate approached 372% in under 24 h. In this framework, hydroxyl radicals ($\text{OH}^\bullet$) were the main ROS that caused the destruction of algal cells.

ROS are also spontaneously generated in aerobic organisms as byproducts of oxidative metabolism in the cytosol, mitochondria, peroxisomes, plasma membranes, and chloroplasts (Almeida et al. 2017; Gill and Tuteja 2010). Although cellular equilibrium depends on an appropriate level of intracellular ROS, too much ROS can cause oxidative damage. Because of their function in photosynthesis, chloroplasts in algae are important locations for the production of ROS that is generated via PS I and PS II. The oxygen produced in chloroplasts can take up leaky reactive electrons from the electron transport chain (ETC) when the energy of the received light surpasses the energy needed for photosynthetic processes. Superoxide radicals ($\text{O}_2^\bullet$), the main sources of superoxide in chloroplasts, are created as a result of this mechanism (Gill and Tuteja 2010). After being further dismutated into H_2O_2 , superoxide radicals undergo additional reactions to produce hydroxyl radicals ($\text{OH}^\bullet$). In PS II, singlet oxygen (O_2) is mostly created when oxygen combines with pigment molecules.

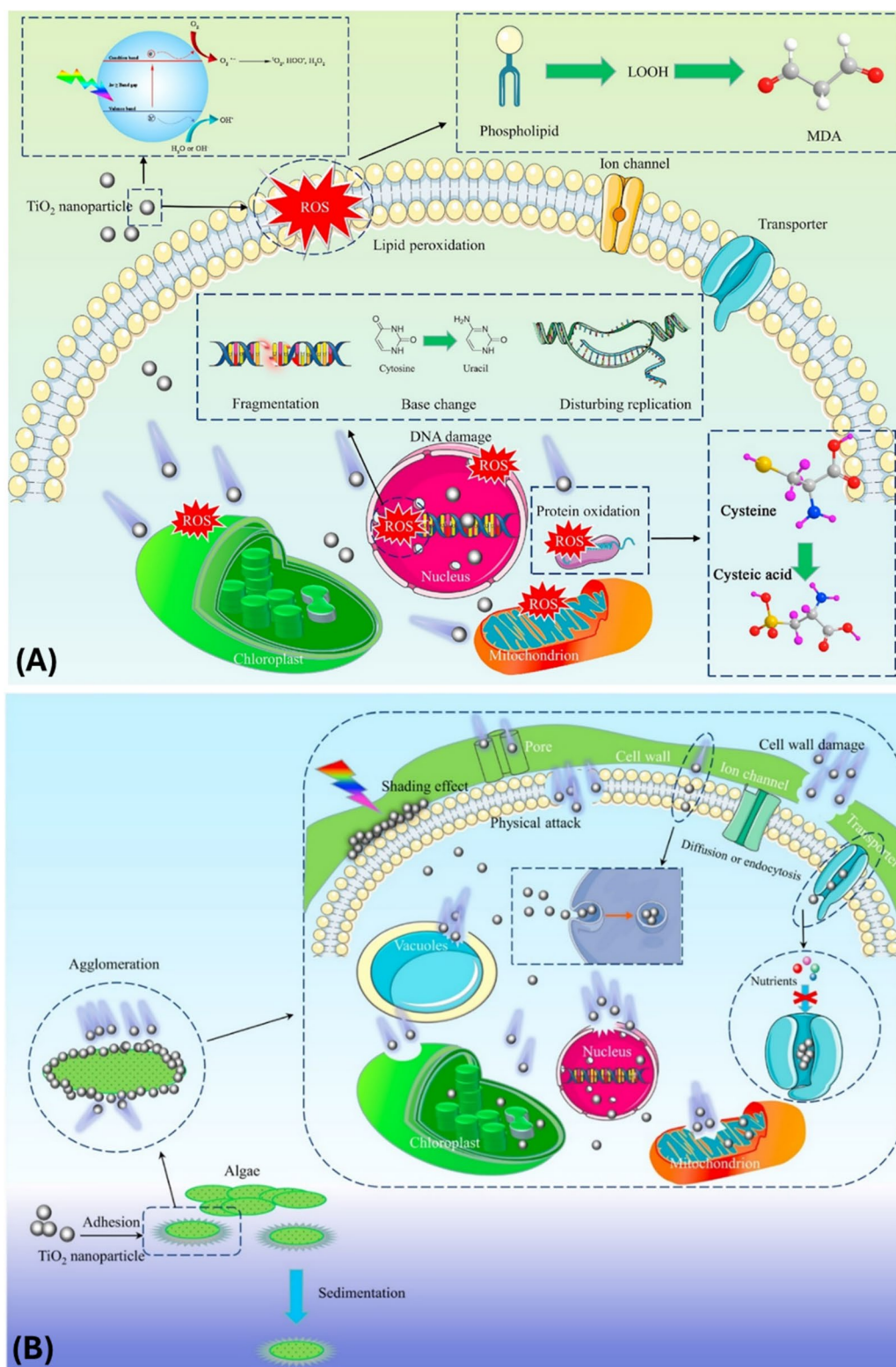
Algae use antioxidant enzymes like catalase (CAT), peroxidase (POD), and superoxide dismutase (SOD) to combat ROS. According to Singh et al. (2018), these enzymes function as markers of oxidative stress. The anti-oxidative defense system can be overpowered by high levels of ROS in algal cells under stressful conditions, such as exposure to biocides, gamma radiation, microplastics, or photocatalytic materials (Malina et al. 2019; Thiagarajan et al. 2019). For example, photocatalytic particles might interfere with the ETC or other metabolic processes, causing an imbalance amid the generation of ROS and the body's defenses against them (Sousa et al. 2018). According to Zhang et al. (2020), oxidative damage is the outcome of this imbalance. Furthermore, the production of photosynthetic pigments can be disrupted, and the structure of photosynthetic systems, including

the D1 protein of the PS II reaction center, is damaged by photocatalysts or the metal ions they produce (Sadiq et al. 2011; Sousa et al. 2018). A decrease in pigments is frequently associated with an excess of intracellular ROS, which inhibits photosynthesis (Hou et al. 2019). Excessive buildup of ROS occurs both intracellularly and extracellularly as a result of photocatalysis and the oxidative stress it causes (Chen et al. 2019).

The emphasis ought to be changed to the toxicological effects of ROS in order to better understand the mechanisms by which they harm algal cells. Lipid peroxidation (LP), DNA and RNA degradation, and protein oxidation are all possible outcomes of ROS-induced damage. Damage to nucleic acids interferes with transcription and cellular replication, whereas LP weakens cell membranes. Structural stability and enzymatic activity may be compromised by protein oxidation. The severe impact of ROS on algal cells is a result of several toxicological effects taken together, underscoring the necessity of more research into these pathways.

5.2.1.1 Lipid peroxidation The process of LP, which happens during regular aerobic metabolism, turns toxic when biological systems experience an overabundance of ROS. The phospholipid bilayer and protein transporters that make up the cell membrane of algae function as a vital barrier, controlling the exchange of nutrients and metabolites and protecting the organism from external stresses (Gerken et al. 2013). Malondialdehyde (MDA), an indicator for oxidative damage, is produced when lipid hydroperoxides (LOOH) are formed when ROS extract hydrogen from the unsaturated chains of phospholipids during lipid peroxidation (Rezayian et al. 2019). Elevated permeability, reduced selectivity, fluidity, and integrity are the outcomes of ROS-induced membrane degradation, which allows intracellular metabolites, small molecules, and ions to leak out (Ghezzi and Bonetto 2003). Photocatalyst particles can enter cells and break down biological components because of this injury (Fig. 10). Rich in polyunsaturated fatty acids, chloroplast, and mitochondrial membranes are especially susceptible to ROS attack, which can result in LP and functional degradation (Nadtochenko et al. 2005).

Algae have shown notable responses to photocatalytic compounds. When exposed to visible light for four hours, $\text{Bi}_2\text{Mo}_3\text{O}_{12}$ nano-foam photocatalysts



◀**Fig. 11 A** Schematic representation of oxidative stress induced via the ROS formation (reprinted with permission from He et al. (2021)). **B** Schematic representation depicting the physical restraint effects of TiO₂ on algal cells (reprinted with permission from He et al. (2021))

demonstrated an 84% photodegradation efficiency for chl-a in *M. aeruginosa*, with elevated MDA and glutathione levels suggesting ROS-induced lipid peroxidation (Li et al. 2018). Interestingly, O₂^{•-} was found to be the main ROS in charge of degrading algal cells. Similarly, after 8 h of exposure, *Chlamydomonas reinhardtii* showed significant MDA accumulation, indicating lipid oxidation, due to nano-TiO₂ (Von Moos and Slaveykova 2014). Lipid peroxidation brought on by oxidative stress resulted in notable growth suppression at high TiO₂ concentrations (10 mg/L or more) (Chen et al. 2012). Under visible light, 90.4% of *M. aeruginosa* was eliminated by a Cu₂O-montmorillonite photocatalyst in 180 min, and elevated MDA levels were associated with membrane lipid peroxidation (Gu et al. 2016). Furthermore, Roy et al. (Roy et al. 2018) found that *Scenedesmus obliquus* exposed to UVA-irradiated nano-TiO₂ had elevated ROS and MDA levels, that resulted in lipid peroxidation and membrane damage. These results highlight how important ROS-induced lipid peroxidation is in photocatalysis-induced algal cell damage.

5.2.1.2 DNA damage ROS attacks primarily target nucleic acids at the cellular level. According to Singh et al. (2009), photocatalytic particles have the ability to reach the nucleus and cause a surplus of ROS, that may negatively impact proteins, lipids, and nucleic acids. Increased ROS levels degrade cellular activities by interfering with vital cellular processes like transcription, translation, and gene replication. The most harmful ROS is OH[•], which targets every part of DNA, resulting in damage to nucleobases and saccharides, sugar-phosphate backbone breakage, and single- and double-strand breaks (Nadtochenko et al. 2005). O₂^{•-} and H₂O₂ have demonstrated negligible direct harm to DNA, whereas ¹O₂ mainly targets guanine (Gill and Tuteja 2010). Growth and metabolism can be greatly impacted by ROS-induced changes in DNA bases, which can lead to deletions, mutations, and structural instability (Fig. 11A). These

changes can also cause membrane damage, decrease protein synthesis, and denature photosynthetic proteins. The basic mechanism for HAB inactivation is via ROS generated by the photocatalysts that disrupt the cell membrane, chl-a, metabolic activities, etc. The performance of photocatalysts varies via exposure time, light sources, doses, etc. (He et al. 2021). Table 9 represents the inactivation efficiency by different varieties of photocatalysts for HAB treatment. Across all the different classes of photocatalysts for HAB treatment -OH contributes 40–60% while •O₂⁻ (superoxide radicals) that perform the oxidative damage to algal lipids and proteins, contributes 30–40% under the visible light (Yang et al. 2023b; Song et al. 2023) (Table 10).

Additionally, ROS affects the generation of toxins in hazardous algal blooms (HABs). Genes include mcyA, mcyB, mcyD, and mcyH control the production of microcystin in cyanobacteria (Valko et al. 2006). Exposure to photocatalysts may interfere with the expression of these genes, which could change the synthesis of microcystin. It's interesting to note that stressors including exposure to nitrite or glufosinate have been demonstrated to upregulate these genes, which increases the synthesis of microcystin (Chen et al. 2011; Zhang et al. 2017). Even while interactions between nanoparticles and algae are receiving more attention, little is known about the cytotoxic effects of DNA/RNA damage brought on by nanoparticles, which calls for more research.

5.2.1.3 Protein oxidation By oxidizing amino acid side chains, breaking peptide bonds, and forming disulfide bonds between sulfur-containing amino acids, ROS can seriously impair the structure and functionality of proteins (Rezayian et al. 2019). Changes in electrical charges and protein cross-linking are other consequences of these attacks. Since most protein oxidation cannot be reversed, oxidized proteins are more susceptible to proteasomal complex breakdown (Ghezzi and Bonetto 2003). Particularly vulnerable to ROS are proteins with thiol groups, such cysteine and methionine, and sulfur-containing amino acids (Rezayian et al. 2019). These residues are particularly targeted by ¹O₂ and OH[•], which interfere with essential protein functions (Gill and Tuteja 2010). Serious metabolic problems are brought on by ROS-induced

Table 9 Inactivation efficiency by different varieties of photocatalysts for HAB treatment (He et al. 2021; Yang et al. 2023a, b, c; Song et al. 2023)

Photocatalyst type	Inactivation efficiency (%)	Time (h)	Dose (mg/L)	Light
TiO ₂ -based	90–99%	3–32	100–4000	UV–visible
g-C ₃ N ₄ composites	74–94%	3–6	6–2000	visible
MOFs	93–99%	3–6	10–60	visible
LDH-based	80–95%	2–4	50–100	visible
Graphene/GO-based (Z-scheme)	92–98%	4–6	50–200	visible
COFs	85–95%	3–6	20–100	visible
Biochar	75–90%	6–12	100–500	visible
Zeolites composites	80–92%	4–8	100–300	UV–visible

protein oxidation in important organelles, viz., nucleus, mitochondria, and chloroplasts, which hinder electron transport, DNA synthesis, and repair. H₂O₂ and ¹O₂, which prevent the production of proteins in thylakoid membranes and hinder photosystem repair operations, particularly affect chloroplast proteins (Allakhverdiev and Murata 2004). Fan et al. (2019c) immobilized Ag/AgCl@ZIF-8 on a sponge modified with sodium dodecyl benzene sulfonate to create a floating photocatalyst. In *M. aeruginosa*, this composite showed 99% chlorophyll-a removal efficiency after 6 h of photocatalytic reaction under visible light. Significant protein oxidation and organelle damage brought on by ROS were indicated by the algal cells' total soluble protein content dropping to 15.3% of its starting level. O₂^{•−} was the main ROS that caused this damage, followed by OH[•] and h⁺.

5.2.2 Physical restraints

It is commonly known that nano-photocatalysts inhibit the development and metabolism of algae through physical constraint effects. Hetero-agglomeration, the process by which photocatalytic nanoparticles adhere or adsorb to algal cells, is the first of these effects. Agglomerates occur on the algal surface as a result of interactions between nanoparticles and the cell wall and membrane (Zhao et al. 2014). The composition and ultrastructure of the cell wall, including the polysaccharides, uranic acids, and glycoprotein matrix, which promote the adhesion of nanoparticles, are crucial for this interaction (Chen et al. 2019). According to Sendra et al. (2017b), the presence of negatively charged sites

Table 10 Mechanism of removal of *M. aeruginosa* by different photocatalysts

Photocatalyst	Removal mechanism	References
Ag ₂ O/g-C ₃ N ₄ hydrogel	Synergistic adsorption and photocatalysis	Fan et al. (2021b)
AP-EGC-CT	Synergistic adsorption and photocatalysis	Wang et al. (2018b)
Ag ₃ PO ₄ /g-C ₃ N ₄	Radical ·O ^{2−}	Sun et al. (2022b)
g-C ₃ N ₄ /Bi-TiO ₂	Active oxygen h ⁺ and ·OH	Song et al. (2021)
A-GUN-coated foam	Hydroxyl radical (·OH) and hole (h ⁺)	Fan et al. (2021c)
Biochar-supported nZVI (BC/nZVI)	·OH, ¹ O ₂ , ·O ₂ [−] (Fenton-driven)	Wei et al. (2025)
g-C ₃ N ₄ @Bi ₂ MoO ₆ @AgI	·O ₂ [−]	Sun et al. (2022a)
g-C ₃ N ₄ -MoO ₃	h ⁺ , ·O ₂ [−] and ·OH	Wang et al. (2020)
Floating g-C ₃ N ₄	Radical ·O ^{2−} and ·OH	Song et al. (2018)
ZnFe ₂ O ₄ /Ag ₃ PO ₄ /g-C ₃ N ₄	Radical ·O ^{2−} and ·OH	Fan et al. (2022b)
g-C ₃ N ₄ /calcite/pumice floating photocatalyst	·O ₂ [−] and ·OH	Zhang et al. (2025a)
TiO ₂ /MIL-100(Fe) S-scheme heterojunction	·OH, ·O ₂ [−] , ¹ O ₂	Zhang et al. (2025b)
Sulfuric acid exfoliated C ₃ N ₄	h ⁺	Wang et al. (2022)

on the cell wall further increases attraction to positively charged nanoparticles.

Adhesion of nanoparticles frequently results in shadowing effects, which limit the amount of light that may reach photosynthetic centers and impede photosynthesis. By weakening cell walls, producing deformation, and jeopardizing membrane integrity, fluidity, and selectivity, homoagglomeration-the clustering of nanoparticles-exacerbates these consequences (Wang et al. 2016). Nanoparticles can cause plasmolysis and organelle destruction once they have broken via the cell wall and entered the cytoplasm (Xia et al. 2015). The nanoparticles may diffuse into the cell by endocytosis or embedded transport proteins if the pore size of the cell wall is larger than the nanoparticles (Navarro et al. 2008). The sedimentation effect is further exacerbated by particle aggregation, which makes algal cells heavier and may cause them to sink below the water's surface where there is inadequate light (Fig. 11B).

Additionally, by trapping cells or obstructing transporters, photocatalysts limit the absorption of nutrients that are essential for HAB growth, such as nitrogen, phosphorus, and vitamins. Fan et al. (2018) examined the effects of five MOFs on the growth inhibition of *M. aeruginosa*. The order of inhibitory activity was Cu-MOF-74 > Ag/AgCl@ZIF-8 > ZIF-8 > Zn-MOF-74 > MIL-125(Ti), with Cu-MOF-74 exhibiting notable membrane damage at 1 mg/L. According to Wang et al. (2016), nano-TiO₂ prevented *Phaeodactylum tricornutum* from growing by physically rupturing cell walls instead of causing oxidative stress. Similarly, Armelao et al. (2009) discovered that *Pseudokirchneriella subcapitata* was caught in aggregates of nano-TiO₂, which inhibited its growth. These results highlight how crucial physical interactions are to nano-photocatalyst toxicity mechanisms.

6 Factors affecting the photocatalytic algal inactivation

Numerous aspects play an important function in photocatalysis, prompting the catalyst's effectiveness both synergistically or antagonistically. Essential factors altering algal inactivation are reviewed in the subsequent segment.

6.1 Initial HAB cell count

Pollutant concentration and type considerably influence the competence of the photocatalytic process. Numerous research (Cao et al. 2024; Fan et al. 2023, 2021b, 2021c) have investigated how preliminary algal cell density alters adsorption-desorption equilibrium. Extremely low or excessive concentrations decrease active spots and reduce ROS effects, reducing effectiveness. Wang et al. (2018c) observed 98% algal deactivation at exposure hours beyond 8 h. Fan et al. (2020c) noticed higher *M. aeruginosa* deactivation (99%) at 6.48×10^6 cells/mL, whereas smaller concentrations (2.16×10^6 and 4.32×10^6 cells/mL) lead to 88% and 92% effectiveness, respectively. At elevated concentrations of algal cells, higher inhibition was observed suggesting an increased interaction between cells and catalysts. Wang et al. (2008) reported that at very low algal cell concentration (0.9×10^6 cells/mL) catalyst-cell interactions was hindered, but at 5.2×10^6 cells/mL density 99% degradation was observed in 9 h. An optimum concentration (2.7×10^6 cells/mL) attained substantial results in 3 h, indicating the significance of maintaining equilibrium cell density for improved photocatalytic effectiveness.

6.2 Photocatalytic dosage

Catalytic dosage plays an imperative task in photocatalytic systems besides the initial contaminant concentration. Various dose-dependent experiments on algal deactivation (Fan et al. 2022a; Long et al. 2022; Yu et al. 2023) indicate that tremendously elevated or minimal dosages do not warrant excellent results. Though higher amounts enhance active spots and ROS production, extreme catalyst load can initiate light scattering owing to solution turbidity, lowering penetration effectiveness. Furthermore, at extreme dosages, the agglomeration, flocculation, and precipitation reduce the amount of existing active sites. Yu et al. (2023) established that raising the dosage from 0.1–0.4 g/L increased *M. aeruginosa* deactivation from 32% to 99.04%. Likewise, Fan et al. (2020c) noted increased algal inactivation when the dosage was augmented from 0 to 200 mg/L; however, at 250 mg/L, effectiveness was reduced caused by agglomeration, thus decreasing the ROS production. Further, Fan et al. (2021d) studied the influence of catalyst load on A-GUN@SMF composites,

observing that A-GUN-SMF-1 displayed excellent findings with 98% inhibition in 3 h. Nevertheless, at reduced (0.5 g/L) or high (2 g/L, 3 g/L) dosage, the efficacy declined due to inadequate ROS production and too much surface saturation, restraining visible light utilization.

6.3 Type of catalyst and HAB strain

Several MOFs have been produced for the inactivation of *M. aeruginosa*, using Cu-MOF-74, ZIF-8, Zn-MOF-74, etc. (Fan et al. 2018). These MOFs have clearly shown their steadiness, alterations in surface area due to organic ligands, penetration potential, and mode of action. At 1 mg/L, Cu-MOF-74 reduced algal growth via agglomeration of the catalyst on cell surfaces and membrane outflow. Yu et al. (2023) assessed the influence of catalysts on various algal strains in visible light-driven (VLD) photocatalysis, observing 66.84%, 91.06%, and 93.45% inhibition efficiencies for *Nodularia*, *Oscillatoria*, and *Nostoc*, respectively. Differences were associated with variations in morphology and adaptation (Perillo et al. 2022). The co-existing algal strains affected catalyst performance because of hydrophobic and electrostatic interactions influencing agglomeration (Cao et al. 2021). Owing to better surface hydrophobicity and minimal catalyst surface potential, CuO-doped g-CN inactivated *C. vulgaris* more efficiently than *M. aeruginosa*. Furthermore, secondary metabolites can interrupt catalytic activity. Compared to *M. aeruginosa* and microcystins, *C. raciborskii* presented excellent resistance to photocatalytic inactivation (Jin et al. 2019). Algal organic matter (AOM), including intracellular and extracellular components, impedes ROS generation by blocking active sites, reducing VLD photocatalytic efficacy (Diao et al. 2019).

6.4 Agglomeration and shading effect

Two important mechanistic insights of anti-algal action include: (i) adsorptive uptake of algae via the catalyst, causing diffusion and metabolic inhibition, and (ii) ROS generation via visible-light-active (VLA) photocatalysts, instigating leakage and structural disturbances (Rashid et al. 2022). Adsorptive uptake commonly includes algal cell aggregation

nearby catalyst active sites or nanoparticle clumping across algal cells; this process is known as hetero-agglomeration or hetero-aggregation, which breaks external cell layers (Sendra et al. 2017a). This incident is affected by algal surface compounds such as glycoproteins, uronic acids, and polysaccharides, together with negatively charged functional groups (Chen et al. 2019). At variable zeta or electric potentials or elevated catalyst amounts, catalytic agents experience homo-agglomeration, where they cluster in the reaction mixture or inside distressed cells near cytoplasmic components. *M. aeruginosa* treated with Cu-MOF-74 (1 mg/L) forms clusters during VLD photocatalysis due to decreased zeta potential from structural impairment (Fan et al. 2019a). Likewise, nano-CuO (80 mg/L) interrupted *C. vulgaris* (Wang 2013), and nano-SiO₂ stuck to *Scenedesmus*, initiating plasmolysis and cluster formation (Wei et al. 2010). As homoagglomeration can stimulate a shading effect, stopping light invasion into algal cells (Lin et al. 2012), some investigations indicate that prior structural impairment reduces this impact insignificant. Moreover, hetero-agglomeration improves cell weight, initiating HABs to decline, causing nutrient deprivation and interrupted metabolic transport, eventually restraining growth (Hartmann et al. 2010).

6.5 Effect of pH

The pH plays a vital part in photocatalytic processes, equally affecting the physicochemical properties of the catalyst and contaminant (Rashid et al. 2020, 2019). The point of zero charges (pHPZC) influences the surface charge of the catalyst (Rashid et al. 2020). As cell surfaces of algae contain negatively charge (Zhou et al. 2020), a somewhat acidic environment commonly favors VLD algal elimination and mineralization. But this varies depending on the catalyst's pHPZC, which falls between 4.0 and 7.0 for several (Long et al. 2022). (Rashid et al. 2022) observed that at alkaline pH, *BiVO₄@HMT* displayed greater inhibition, in the order pH 9 > pH 7 > pH 5, associated with algae's carbon capture mechanism (CCM) and proton motive force (PMF) activation. On the other hand, improved algal inactivation was observed in acidic conditions exploiting *Ag/AgCl@g-C₃N₄@UIO-66(NH₂)* (pHPZC=6.6) and *A-GUN/SMF* (pHPZC=6.3) (Baniamerian et al. 2020; Fan et al. 2023, 2021a, 2021b, 2020b, 2020a), where

electrostatic interactions accelerated uptake and oxidative destruction (Vega et al. 2018).

6.6 Effect of light and alkalinity

During the photocatalytic technique, the light source is necessary, with experiments employing UV, sunlight, and virtual visible light (Fan et al. 2021c; Rashid et al. 2020). Under various lighting conditions, algal inactivation was minimum in the dark in the absence of A-GUN/SMF catalyst (Fan et al. 2021c). The lack of light stopped photosystem activity, and catalyst addition in dark conditions rendered it unsuccessful. But, in light, substantial algal inactivation was attained within 3 h. A 99% algal inactivation was observed under solar irradiation owing to elevated photon energy and modest heating in contrast to simulated visible light. Phukan et al. (2018) exhibited UV-C light's efficacy, affecting *Nostoc muscorum Meg1* filament damage and structural impairment to RuBisCO, thylakoids, and enzymes. To evaluate the feasibility of VLD algal removal, the natural water environment must be considered. Aspects such as inorganic salts, ions, and ROS scavengers impede photocatalysis. Furthermore, in seawater, the penetration of light is reduced. Fan et al. (2023) monitored the influence of alkaline conditions on Ag_2CO_3 -GO and $Ag_2MoO_4/TACN@LF$ floating catalysts. They observed that raising Na_2CO_3 (0–10 mg/L) concentration, MC-LR, and *M. aeruginosa* inactivation was delayed because of hydroxyl radical ($\bullet OH$) scavenging by HCO_3^- and CO_3^{2-} , developing fewer reactive species ($CO_3^{\cdot -}$ and $OH^{\cdot}$).

6.7 Effect of NOM

Natural organic matter (NOM), specifically algal organic matter (AOM), is a main restraint during wastewater treatment owing to its hydrophobic nature and link with taste and odor concerns (Yang et al. 2023a, b, c). NOM, involving humic acids (HA), fulvic acids (FA), proteins, and polysaccharides, delays photocatalytic algal degradation via three crucial mechanisms. Primarily, elevated NOM concentrations boost turbidness, lowering light penetration and delaying photocatalyst degradation (Ye et al. 2019). Subsequently, NOM competes for active sites present on the catalyst, restricting the adsorption of contaminants. Lastly, ROS are quenched by NOM,

reducing photocatalytic effectiveness (Sun et al. 2023a, b). Experiments have demonstrated that HA and FA affect photocatalytic effectiveness in different ways. For instance, Ag_2CO_3 -GO accomplished 100% degradation in 2 h at 5 mg/L HA and degradation was reduced (20%) when the concentration rises to 10 mg/L, primarily due to the quenching of ROS (Fan et al. 2020a). FA demonstrated greater repressing effects than HA due to its higher aromaticity (Awfa et al. 2020). Conversely, 3D $Ag_2O/g-C_3N_4$ /hydrogel photocatalyst was efficient regardless of HA existence, even though compelling long irradiation and higher amounts (Fan et al. 2021b, 2020c). Moreover, inorganic ions (Cu^{2+} , Ag^+ , Fe^{2+}) increase membrane leak and inhibition, whereas Zn^{2+} was less efficient (Yu et al. 2023). Proper H_2O_2 concentration (1–30 mmol/L) improves deactivation, but > 1 mg/L decreases photoactivity due to ROS quenching and light attenuation (Fan et al. 2020c).

7 Significance and toxicity of post-photocatalytic transformed products

During the photocatalytic inactivation process, ruptured algae cells discharge organic matter and algal toxin into the surrounding system, which provokes contamination in the system. Therefore, it is vital to study the concurrent elimination of organic matter and toxins during the HABs photocatalytic inactivation.

7.1 Degradation of algal organic matters

During photosynthesis, algae incessantly generate algal organic matter (AOM) for their normal growth. Generally, AOM comprises of extracellular organic matter (EOM) which is discharged into the water during metabolism and reproduction, while the intracellular organic matter (IOM) is deposited inside the cells. EOM is moderately adsorbed on the surface of algal cells imparting a negative charge to the surface, enabling cells to be dispersed stably in aquatic system (Yan et al. 2022a, b). Further, IOM comprehends odor-producing compounds, proteins, and algal toxins which are preserved inside the cells but are discharged when cell membranes break. The discharge of AOM not only magnifies organic contamination but also presents problems for water treatment, since

these are more hydrophilic and nitrogen-rich than typical natural organic matter (NOM). Hence, they have a more inclination to establish lethal nitrogenous disinfection by-products during chlorination (Li et al. 2012).

Since photocatalytic deactivation of algae ruptures their cell membranes via ROS, it is crucial to understand whether photocatalysts have the capability to destroy the AOM. Algal cells sustain a negative charge principally because of dissolved organic matter (DOM) adsorbed on their surfaces (Neilen et al. 2017). Shifts in the zeta potential of algal suspensions; therefore, offer understanding into the fate of AOM during photocatalysis. Several researchers have observed that during the photocatalytic process the absolute value of zeta potential of algae suspension increased slightly and later reduced substantially (Fan et al. 2020d, b, a, 2021a). They reported that the algae cells break under the ROS, and the organic matters present internally were discharged and attached to the algae cells surface. Thus, the attachment of negatively charged organic matters enhanced the absolute value of the algae cells (Pei et al. 2016). Since ROS was produced continuously by the photocatalyst, the excessive ROS not only destroyed the cell wall but also destroyed the organic matters. Therefore, it can be assumed that photocatalysts can not only deactivate algae cells but also destroy the released organic matters, escaping possible water pollution.

Likewise, Yu et al. (2024) used carbon nitride nano-sheets (SCN), and a ternary Z-scheme photocatalyst was constructed via heterojunction engineering for photocatalytic elimination of harmful algae and their organic matter. They observed approximately 100% removal of chlorophyll in 300 min. Three-dimensional fluorescence spectrograms and cellular SEM maps showed elimination of algae and their organic matter. Mechanistic studies displayed that surplus $O_2^{\cdot-}$ played a critical role in algal cell deactivation and organic matter degradation. Li et al. (2026) studied the self-floating carbon-fiber cloth deposited with Ag/Ag_3PO_4 photocatalyst for effective elimination of algal cells and organic matter. Likewise, IOM spectra confirmed characteristic peaks (A-D), where peak B corresponds to the aromatic substances, which slightly decreased in control but showed a sharp decrease under photocatalysis, representing severe membrane damage and loss of metabolic activity. Total organic carbon analysis also confirmed strong

mineralization of released organics by S-AP/CC. Overall, the photocatalyst efficiently deactivated algae and destroyed released AOM, decreasing secondary contamination. These results together prove that photocatalytic processes can eliminate algae while concurrently destroying both EOM and IOM, lowering the threat of secondary pollution.

7.2 Degradation of microcystin

During HABs, enormous amounts of microcystin were released by *M. aeruginosa* into surrounding waters because of cellular burst. Amongst these toxins, MC-LR is the extremely widespread and lethal, representing about 46–99.8% of total MCs identified in natural waterbodies (He et al. 2016). MC-LR is a cyclic heptapeptide which is depicted by an Adda side chain, with leucine and arginine filling varying sides (Liu et al. 2022b). Due to the cyclic peptide structure and hydrophobic Adda significant chemical strength has been observed, compelling MC-LR resistant to biodegradation and environmental intervention (Cai et al. 2022). Additionally, its hydrophobicity reduces the adsorption of MC-LR onto sediments, granting MC-LR to endure in the water (Albuquerque et al. 2022). Studies have shown that the Adda side chain is vital for hepatotoxicity of MC-LR (Sun et al. 2022a).

However, during the process MCs are moderately mineralized, thus it is crucial to assess the harmfulness of the degradation products. Numerous toxicity studies displayed that the MC-LR degradation products do not hinder the *D. magna* growth (Khadgi and Upreti 2019; Sun et al. 2023b). A comet assay utilizing the mouse lymphocytes displayed that the tail DNA, tail length, and tail moment reduced severely with the degradation of MC-LR (Zhao et al. 2019). At the molecular level, the estimation of protein phosphatase inhibition by MC-LR degradation products exhibited that the toxicity of MC-LR was completely linked to the MC-LR presence and that the degradation products did not impact on the enzyme function (Fotiou et al. 2013; Choi et al. 2021). These studies may be accredited to the aromatic ring hydroxylation of the conjugated diene or Adda side chain efficiently decreasing the MC-LR toxicity. But molecular toxicity tests indicated that the toxicity of some degradation products is greater than that of MC-LR (Zong et al. 2013).

Table 11 Technique for separating the photocatalysts after post-HAB treatment from the water bodies (Song et al. 2023)

Photocatalyst	Separation method	Advantages	Limitations
Floating	Skimming	Easy recovery, sunlight availability	Wind dispersion in large areas
Powdered	Sedimentation/Filtration	High activity	Poor recovery, might impose secondary pollution
Magnetic Composites	Magnetic extraction	High recyclability, low loss	Additional synthesis cost

Ouyang et al. (2023) also assessed the environmental effect of MC-LR intermediate by D-CN/MoO₃ composites photocatalytic degradation. The toxicological and ecotoxicological assessments of the MC-LR intermediates were executed utilizing the Toxicity Estimation Software Tool (T.E.S.T). Results showed that most of the MC-LR intermediates have lower acute and sub-lethal impact in *D. magna* and oral rats compared to the parent MC-LR. However, some intermediates were more lethal and mutagenic, the ultimate photocatalytic mixture was less noxious. Compared to all the pathway intermediates of pathway C of MC-LR photodegradation were found to be less poisonous, suggesting that the D-CN/MoO₃ efficiently degraded MC-LR toxicity.

7.3 Effect of metal leaching from the photocatalysts

Mansfield et al. (2015) assessed the TiO₂/carbon nano-hybrids acute toxicity towards *D. magna*. Concentration-dependent mortality was connected to ROS-mediated oxidative stress due to the accumulation of nanoparticles in the gut, emphasising the possible freshwater jeopardies and the necessity for chronic noxiousness evaluation. Li et al. (2014) studied the toxicity of nano-TiO₂, graphene, and graphene nano-platelets (GNP) in *D. magna* and *Oryzias latipes*. Results revealed that nanoparticles agglomerated, sedimented, and adhered to the surfaces of organisms. Nano-TiO₂ and GNP displayed analogous phototoxicity, whereas GNP produced 2.3-folds more visible-light-induced ROS, signifying the importance of illumination, particle behaviour, and species specificity. Priyadharsan et al. (2024) reported that zebrafish embryos displayed dose-dependent harmfulness after 48 h contact with rGO/g-C₃N₄/FeTiO₃ ternary nanocomposites (6.25–100 mg/L). The hatching rate of embryo was significantly decreased at higher concentrations, although total toxicity persisted to be low, supporting the nanocomposites'

promising capability for photocatalytic and biological systems applications.

8 Future perspectives and outlook

Even though significant research and innovations have been made, still there are many techniques yet to be explored in combination with the on-going remedial approaches. The inclusion of genetic detection along with preceding monitoring techniques, combining the data with the ML models, investigating more toxic production with the HAB's dynamics under different conditions, exploring more efficient composite sustainable materials in combination with the current technologies can be further considered. ML assist in material screening and designing for photocatalysts, optimizing and controlling the HAB treatment operations, kinetic modeling and mechanistic inference of the algal and their toxin degradation. Various surrogate models (e.g., gradient boosting, random forests, GNNs) are being set on trials for mapping the catalyst descriptors and optimized operational conditions for HAB degradation, chl-reduction, cellular inactivation, etc. Also, data driven models (CatBoost, XGBoost, ANN) catalyst loading, HRT, pH, DOC, turbidity, nutrient levels, and initial algal/toxin concentrations as input and removal efficiency, energy consumption, by product formation, etc. as output. These models can further be used as "digital twins" for multi-objective optimization and real-time controlling of the deployed systems.

The building blocks of photocatalysts should comprise more green solvents and be more sustainable. The potential architectonics should be more economical and scalable, and the synthesis process should be continuous/semi-continuous for the on-site application/treatment. In addition, the photocatalyst should be more selective, stable, and long-term durable, along with proper embracing of the design that can

only mitigate the HABs and should not at all harm or disturb any microhabitats. The conjunction of functional composite materials with photocatalysts such as MOFs, COFs, $g\text{-C}_3\text{N}_4$, graphene, zeolites, biochar, etc. that might result in an effective combination of the required properties. On-site research is essential for comprehending the impact of tough climatic conditions on their efficiency, inhibition, and stability of photocatalytic architecture over time. Further studies on the on-site interactions between photocatalysts and HABs should be performed in more detail to determine more mechanistic insights and aftereffects of photocatalysts within the aqueous ecosystem at micro-levels.

Also, absolutely eco-friendly and green photocatalysts still need to be explored, along with proper environmental hazard assessment on a priority basis. The integrated approach comprising photocatalysts along with other removal techniques may be a better option for such an approach that will further lead to opening of new dimensions and opportunities for new hybrid photocatalysts/approaches. Further, the upcycling of photocatalysts can only be possible via recycling, and proper disposal of spent materials is also vital for further research. Also, the customization/tailoring of photocatalysts in accordance with different environmental conditions is crucial. Future investigations should also explore long-term monitoring and determine the ecological consequences of photocatalysts for evaluating their sustainability. The challenges imposed by climatic conditions such as high temperatures, altitudes, precipitation patterns, and extreme weather/calamities should also be taken care of.

Post-HAB treatment, the filtration/magnetic separation is capable of recovering >90% of the photocatalysts, further permitting their reutilization in numerous cycles without any substantial loss. In case of non-recoverable concentration, encapsulation is carried out in stable matrices to avoid any leach out of toxic metal/s into the environment. The direct disposal to the landfill sites is strictly discouraged. In addition, the spent photocatalysts (mostly carbonaceous) can be upcycled via pyrolysis, converting them into carbon-sequestering materials for soil amendment or filtration applications, leading to a circular economy approach. Also, vitrification or cementation can be performed that ensures their long-term stability and very least leaching as per the EPA and EU waste directives. Their environmental fate modelling

shows insignificant secondary pollution when the abovementioned techniques are applied.

In addition, the post-HAB treatment using photocatalysts faces a huge challenge owing to the large-scale HAB removal. Apart from the chemical efficacy, photocatalysis technique faces deprived sunlight penetration in the HAB growth water bodies, scalability in huge open water sources and cost-involved in artificial light sources. Further, catalyst aggregation, biofouling and environmental release menace, further enhance the complexity in such dynamic water bodies. Also, the economic feasibility at municipal-level operations is not possible owing to the huge material requirement and maintenance expenses involved. The powdered photocatalysts are very difficult to recover from the post-HAB treatment from the open water bodies, resulting in photocatalyst loss and further posing the toxicity in the aqueous ecosystem. Table 11 represents some of the different techniques for photocatalyst separation from the water-bodies along with their advantages and disadvantages involved. Furthermore, it is predicted that the temperature, precipitation, nutrient levels, and flow rate patterns are projected to alter in the future. There are numerous discrete freshwater/marine water bodies, that will respond adversely with climate changes. Also, these advanced functional photocatalytic nanomaterials possess sufficient amount of environmental and eco-toxicological risks during their application for HAB removal from the aqueous system. For mitigating these concerns, eco-designing can be promoted for selecting stable frameworks along with surface modifications for minimizing the leaching.

Even though there are many limitations, recommendations, and future perspectives for HAB removal based on to the simulated contaminated system, the evidence-based recommendations will directly address the photocatalytic nanomaterials recovery, eco-toxicity, real-water complex interactions, and long-term applicability. The incorporation of multiple end-point assessment for each experiment along with the tracking of HAB's cellular density (individual and under mixed culture conditions), toxin quantification, ROS markers, post-treatment growth, etc. for the real water matrices are utmost required. In addition, inclusive ecotoxicological testing, including chronic exposure studies on non-targeted model organisms (viz., *D.*

magna, zebra fish) in co-occurrence with the HAB should also be selectively quantified before scaling up along with the long-term operations. Regulatory frameworks should provide non-safety evaluations to keep a balanced innovation along with ecosystem protection.

The explicit monitoring of the intermediates/by-products of toxin degradation, cell-lysis, performance evaluation under raw bloom water consisting full solids, bi-carbonates, and DOC is also required. In addition, the end-of-life management, shifting of non-metal based robust photocatalytic nanomaterials possessing low-leachability and aqueous solubility, their regeneration and upcycling routes, chronic toxicity on multiple trophic levels, performing the life cycle assessment for all the different photocatalytic nanomaterials will definitely lean towards the real-water application and long-term fate of the photocatalysts. The integration of smart control, monitoring, and multi-scale modeling via techno-economic analysis, smart in-situ IoT-based sensing coupled with AI/ML models to minimize non-targeted damage, bloom forecasting, lake/pond-scale mass-balance models etc. Further, the incorporation CAPEX/OPEX, disposal cost, cost-involved for the integrated approach viz., ultrasound, flocculation, H_2O_2 should also be considered.

9 Conclusions

Advanced functional photocatalyst-architectonics are currently the budding class of materials that exhibit profuse superior applications. The present review has comprehensively covered and summarized the recent advancements of these functional photocatalysts for HABs removal from freshwater and marine environments. An explicit focus on deciphering the established techniques for HABs detection on the basis of the morphological structure, cytochrome, immunoassay, and nucleic acid technologies was also discussed. The involvement of biological, physical, chemical, and hybrid techniques for HABs inhibition was also covered. The outstanding potential of the functional photocatalytic nano-architectonics, viz. MOFs, TiO_2 -based metal composites, $g-C_3N_4$, COFs, graphene-based materials, zeolites, porous coordination networks, and hybrid composites involving self-assembled

metal ions/clusters, biochar, etc., for HABs inhibition were evaluated. Additionally, the mechanistic insights and factors affecting the photocatalytic-architectonics inactivation were also covered. Overall, this review has explored and described the potential of advanced functional photocatalyst-architectonics for HABs mitigation from aqueous systems, further unraveling the significant potential for the mitigation of global freshwater scarcity and attaining sustainable development goals.

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Declarations

Conflict of interest The authors declare no competing interests.

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